Spatial fiber steering decouples the nonlinear stiffness of composite panels from their linear frequency: mechanism, limits, and manufacturable design

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Abstract

Question. Thin composite panels harden as they vibrate: their resonance drifts upward with amplitude and, under forcing, folds and jumps. We ask whether the *spatial distribution* of fiber orientation, the freedom that automated fiber placement provides, can reduce this hardening without changing the panel's linear frequency or mass. If it can, we ask what mechanism provides the lever, what bounds it, and how it compares with a uniform ply angle and with the stacking sequence.

Hypothesis. The linear frequency and the hardening integrate the same rotated stiffness against two different strain fields of the fundamental: its bending curvature and the in-plane-relaxed membrane stretch it induces. The two differ from point to point in magnitude and principal direction. A fiber field can then take, point by point, the direction that serves both under the frequency constraint (across the stretch, and toward or away from the curvature as the constraint requires), while a uniform layup can only compromise.

Analysis. We make this exact. A local Fourier representation shows that each energy density depends on the fiber angle only through the anisotropic invariants U_2, U_3 and the local principal directions. This predicts no lever for isotropic material, exact invariance of a single-ply beam under any uniform angle, and a pointwise optimality condition whose violation guarantees that steering beats every uniform angle. We test the predictions with a spectral-submanifold reduction of laminated von Kármán beams and plates. The hardening coefficient is exact in the in-plane relaxation and evaluated on a mode-adaptive fundamental with an analytic adjoint. The frequency is imposed as an exact equality and the tow-curvature limit pointwise.

Results. On a clamped beam, symmetric steering with no bending–extension coupling cuts the detuning by up to 59% with the frequency kept within 2.4% of the unidirectional value ($\omega_1 \geq 0.98\,\omega_{1,\mathrm{UD}}$ in the design model; 0.976 on the full model), by placing compliant fibers at the inflection points. An optimized bending–extension coupling, the intuitive alternative, gives only 2% (on its own drift objective). On a simply supported plate, a curvature-feasible tow field reduces the frequency-normalized hardening to 0.366 of the unidirectional value at the unidirectional frequency (held exactly in the design model, to within 1.4% on refined bases and an independent finite-element model). No uniform layup can do this: the unidirectional plate is the lowest-frequency member of all 20,736 symmetric stacking sequences, and the steered designs also undercut the best stacks at every higher frequency tested. A pointwise design computed from the unidirectional fields alone already captures most of the gain, and steering raises the relaxed share of the stretch energy from 33% to 51%. The lever grows monotonically with lamina anisotropy and saturates once the minimum tow radius is below about 20 mm, a fraction of the scale of the strain fields. The full multi-mode model confirms the reduction by its SSM and by EPMC continuation. Direct time integration shows the consequence: the jump-free amplitude widens by 1.66×, as predicted by $a_c \propto J^{-1/2}$, and at a drive where the unidirectional panel jumps with hysteresis, the steered panel is single-valued.

Conclusion. Spatial fiber steering is a genuine, mechanistically understood lever on nonlinear stiffness at fixed linear frequency and mass. It is set by material anisotropy and by the tow

radius relative to the strain-field length scale, and for this plate it is far stronger than the through-thickness freedom of stacking.

Keywords: variable-stiffness composites; fiber steering; geometric nonlinearity; spectral submanifolds; backbone curve; adjoint optimization; automated fiber placement.

1 Introduction

Motivation. Thin composite panels stiffen as they vibrate. Mid-plane stretching makes the restoring force of a bending mode cubic in its amplitude, so the resonance frequency drifts upward with amplitude along the *backbone curve* $\Omega(a)$ [8, 1]. Under harmonic forcing, the hardening resonance leans over, folds, and *jumps* between coexisting branches with hysteresis. For a resonant sensor the drift is a calibration error. For a harvester or an actuator the jump is a loss of function. In practice, the designer of such a panel is not free to move its linear frequency (set by the application) or its mass (set by the vehicle). The design question is therefore whether the *nonlinear* stiffness can be changed *independently* of both.

1.1 Related work and positioning

Reduction and design of nonlinear normal modes. The backbone is the frequency–amplitude relation of a nonlinear normal mode [1, 2, 10]. Spectral submanifold (SSM) theory makes it the reduced dynamics on the unique smoothest invariant manifold tangent to a modal subspace [3]. Its parametrization can be computed directly from finite-element models to arbitrary order [4, 5, 6, 7]; see [8] for a review of geometrically nonlinear reduction. This makes nonlinear response a *designable* quantity. Dou et al. shaped normal-form coefficients of beams and frames by adjoint shape optimization [12, 13]. Li derived explicit SSM sensitivities [14]. Pozzi et al. tailored conservative backbones through the reduced dynamics of Lyapunov subcenter manifolds [16], obtained adjoint sensitivities of SSM backbone curves at arbitrary order [15], and optimized the topology of the hardening/softening coefficient [17]. Liang et al. extended adjoint SSM sensitivities to forced-response curves and density-based topology optimization [18]. Force-based reduced-order models have also been used to optimize nonlinear finite-element models [19]. In all of these the design variables are *shape*, *size* or *material density* of isotropic structures, and the question is how to reach a target response, with the frequency held alongside it by an equality target on the backbone frequency at a prescribed amplitude [15] or a tolerance constraint on the linear frequency [18]. None of them uses a fiber-orientation field, and none imposes the curvature limit of automated fiber placement.

Variable-stiffness composites. Steering tows along curvilinear paths was introduced for buckling and stress concentration [22]. It is now a mature design space for *linear* objectives: in-plane stiffness and buckling [23, 24, 26], and fundamental frequency through lamination parameters [25, 21]. It has also been optimized for a geometrically nonlinear *static* objective, postbuckling stiffness [33, 34], where the gain is attributed to the in-plane redistribution of load toward the supported edges [23, 33]; the relaxed-stretch mechanism identified here is closely analogous to that redistribution. Orientation is also optimized pointwise in topology-optimization frameworks [41, 42], and continuous-fiber printing [43] with toolpath-integrated, curvature-constrained design [44] now makes such fields manufacturable. The *nonlinear* vibration of variable-stiffness plates has been *analyzed*: Akhavan and Ribeiro computed the linear modes [38] and the forced periodic vibrations [27, 28], and Ribeiro the nonlinear free vibrations [39] (see also [40]), of plates with *prescribed*

curvilinear fiber paths, Houmat analyzed nonlinear free vibration with curvilinear fibers [29], and Yan et al. built a semi-analytical framework for parametric studies of nonlinear free and forced response [30]. Farsadi et al. optimized curvilinear fiber paths for the *linear* fundamental frequency and then examined the amplitude-dependent nonlinear frequency of the optimum [31, 32]. To our knowledge the hardening itself has not been the design objective, and the reason why a fiber field can or cannot change it has not been identified.

Why it matters. Designers of resonant devices already exploit or cancel geometric nonlinearity by tuning design parameters, for example by balancing mechanical hardening against electrostatic softening to extend the linear range of nanomechanical sensors [20]. A composite panel offers an additional freedom that leaves geometry, mass and material unchanged. This paper asks whether that freedom acts on the nonlinearity at all, and why.

What is new here. Relative to the SSM-optimization literature, the design variable is a manufacturable fiber-orientation field under the tow-curvature constraint, at fixed mass and an exactly held linear frequency (an equality, as in the backbone targets of [15], but with the material fixed and only the orientation free). Relative to the variable-stiffness literature, the objective is the nonlinear backbone, and the result is not only an optimum but a *mechanism*. A local Fourier theory predicts when steering must help, where the fibers go and what limits the gain, and each prediction is tested against the competing laminate freedoms: a uniform angle and the stacking sequence.

1.2 Question, hypothesis and predictions

Scientific question.

> *For a thin composite panel of fixed mass and fixed linear frequency, can the* spatial distribution *of fiber orientation reduce the geometric hardening of its fundamental mode? If so, what physical mechanism provides the lever, what bounds it, and how does it compare with the classical laminate freedoms — a uniform ply angle and the stacking sequence?*

Hypothesis. (Fig. 1.) The linear frequency and the hardening are integrals of the *same* rotated stiffness tensor against two *different* strain fields of the fundamental mode. The frequency is set by the bending curvature, $\omega_1^2 = \int \boldsymbol{\kappa}^\top \boldsymbol{D}(\theta)\boldsymbol{\kappa}$. The hardening is set by the membrane stretch the mode induces after the in-plane field has relaxed, $\gamma = \frac{1}{2}\int \boldsymbol{\varepsilon}_r^\top \boldsymbol{A}(\theta)\boldsymbol{\varepsilon}_r$ (Sec. 2). The two strain fields differ from point to point in magnitude and in principal direction. We hypothesize that a spatially varying fiber angle can therefore satisfy, *pointwise*, a trade-off between the two energies that a uniform angle can only satisfy *on average*: turn the fibers across the principal stretch where stretching energy dominates, and, where bending energy dominates, set them toward or away from the principal curvature as the frequency constraint requires (away from it when the frequency is held at the unidirectional value, the lowest of all uniform layups). Stacking separates the same two functionals in the classical way, through the thickness ($\boldsymbol{D}$ weights the outer plies, $\boldsymbol{A}$ all plies equally [21]); it is the competing freedom against which the in-plane route is tested. Section 3 turns the hypothesis into exact statements. It makes five falsifiable predictions:

P1 An isotropic material offers no orientation lever at all (Corollary 2).

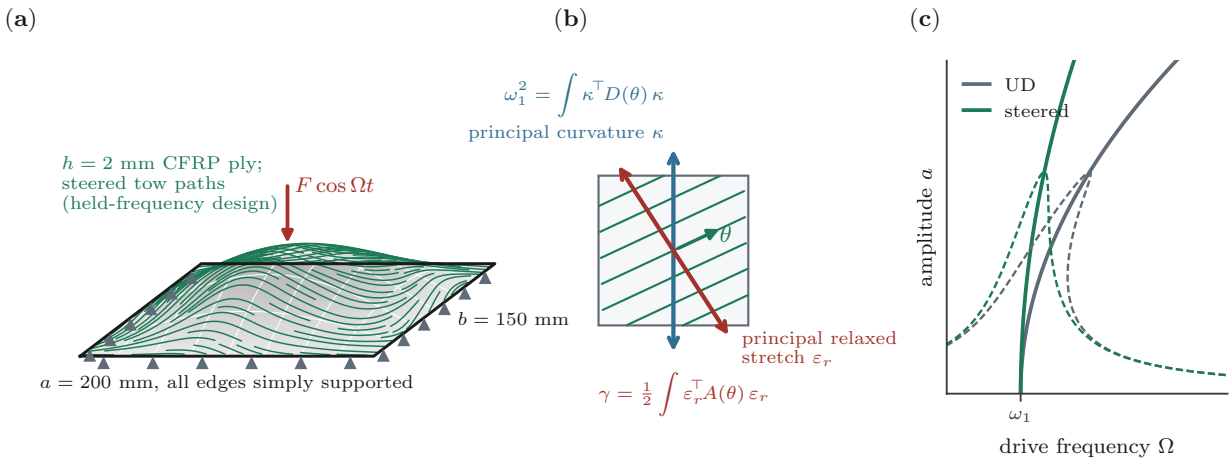


Figure 1: The problem. **(a)** A simply supported $200 \times 150 \times 2\,\mathrm{mm}$ CFRP plate, drawn deflected in its fundamental mode, with the tow paths of the held-frequency steered design of Sec. 5.2 and the harmonic center load. **(b)** What one material point sees: the fiber direction θ, the principal curvature of the mode, which with the bending stiffness $\boldsymbol{D}(\theta)$ sets ω_1^2, and the principal relaxed stretch (drawn at the plate's typical 33° conflict with the curvature, Sec. 5.3), which with the membrane stiffness $\boldsymbol{A}(\theta)$ sets the hardening γ (Sec. 2). **(c)** The consequence, sketched: at the same linear frequency ω_1, a smaller hardening bends the backbone (solid) less, and the forced response (dashed) of the unidirectional plate folds and jumps where the steered one stays single-valued (Sec. 5.6).

P2 For a single-ply beam, any uniform angle leaves the frequency-normalized hardening *exactly* unchanged, so every gain must come from spatial variation. The optimal variation places compliant fibers where the mode does not bend: the inflection points (Proposition 6).

P3 On a plate, a steered field reaches combinations of linear frequency and hardening that no uniform layup reaches, whether a single angle or any stacking sequence.

P4 The optimized orientation follows the local preferred directions of the two energies. A first-order pointwise design computed from the unidirectional mode alone already captures most of the gain (Proposition 5).

P5 The lever grows with the material anisotropy and is limited by the tow-curvature constraint relative to the length scale of the strain fields.

A consequence follows if the predictions hold. The jump-free amplitude scales as $J^{-1/2}$ (Proposition 3), so reducing the hardening at held frequency widens the panel's hysteresis-free operating range by a predictable factor.

1.3 Approach and findings

Approach. We test the predictions on a clamped laminated beam and a simply supported laminated plate, both von Kármán. The backbone is computed by an SSM reduction verified against analytic, harmonic-balance (EPMC) and finite-element references. For design we use the SSM's hardening coefficient in a closed form that is exact for the in-plane relaxation and evaluated on a *mode-adaptive* fundamental, with an analytic adjoint. For the plate the frequency is imposed as an exact equality, so every reported plate gain is at the stated frequency and fixed mass; for the beam

it is a floor ($\omega_1 \geq 0.98\,\omega_{1,\mathrm{UD}}$). The tow-curvature limit is imposed pointwise. The held-frequency plate design is confirmed independently on the full multi-mode model, by its SSM and by EPMC continuation, by direct time integration of up/down frequency sweeps on a six-mode modal model of that plate, and on an independent finite-element model.

Findings in brief. Four predictions are borne out; the mechanism prediction (P4) holds partly.

- **Beam (P1, P2).** Uniform angles leave the hardening invariant to four digits. Symmetric steering, with no bending–extension coupling at all, cuts it by 59% with the frequency within 2.4% of the UD value, by placing compliant fibers at the inflection points. An optimized bending–extension coupling, the intuitive alternative, gives only 2% (on its own drift objective).
- **Plate (P3).** A curvature-feasible steered tow reduces the frequency-normalized hardening to 0.366 of the unidirectional value at its frequency (an exact equality in the design model; within 1.4% on finer bases and on an independent finite-element model). No uniform layup can match this: the unidirectional plate is the lowest-frequency member of all 20,736 symmetric stacks we enumerate. The steered designs also undercut the best stacking sequences at every higher frequency we test.
- **Mechanism (P4).** The pointwise design computed from the UD fields alone, with no optimizer, already gives $J/J_{\mathrm{UD}} = 0.46$ (at $0.935\,\omega_{1,\mathrm{UD}}$); the optimized field shares its large-scale pattern but departs from it by 28° RMS where the energies are large. Steering also raises the relaxed share of the stretch energy from 33% to 51%.
- **Limits (P5).** The lever grows monotonically with anisotropy and saturates once the minimum tow radius is below about 20 mm.
- **Consequence.** The full-order model confirms the reduction (SSM 64%), and the jump-free amplitude widens by 1.66×, as predicted.

Organization. Section 2 states the model, the reduction and the backbone and jump quantities. Section 3 develops the mechanism and its predictions. Section 4 poses the design problem and derives its sensitivities. Section 5 tests P1–P5 and the consequence. Section 6 discusses what the results do and do not establish. The appendices give the SSM recursion, the SSM–condensation equivalence, the finite-element validation, calibration against prior work, and a verification and reproducibility map.

2 Model, reduction, and the quantities we design

2.1 Laminate constitutive model

A unidirectional CFRP lamina ($E_1 = 131\,\mathrm{GPa}$, $E_2 = 9\,\mathrm{GPa}$, $G_{12} = 5\,\mathrm{GPa}$, $\nu_{12} = 0.27$, $\rho = 1600\,\mathrm{kg/m^3}$) has the on-axis plane-stress stiffness $\boldsymbol{Q}$, with $Q_{11} = E_1/(1-\nu_{12}\nu_{21})$, $Q_{22} = E_2/(1-\nu_{12}\nu_{21})$, $Q_{12} = \nu_{12}Q_{22}$, $Q_{66} = G_{12}$ and $\nu_{21} = \nu_{12}E_2/E_1$. In engineering Voigt notation (strain $\boldsymbol{s} = [s_{xx}, s_{yy}, 2s_{xy}]^\top$), rotating the fibers through θ gives

$$\bar{\boldsymbol{Q}}(\theta) = \boldsymbol{T}(\theta)\,\boldsymbol{Q}\,\boldsymbol{T}(\theta)^\top, \qquad \boldsymbol{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \quad (c,s) = (\cos\theta, \sin\theta). \tag{1}$$

Its entries are trigonometric polynomials of degree four in θ (Tsai–Pagano [21]):

$$\bar{\boldsymbol{Q}}(\theta) = \boldsymbol{\Gamma}_0 + \cos 2\theta\, \boldsymbol{\Gamma}_1 + \sin 2\theta\, \boldsymbol{\Gamma}_2 + \cos 4\theta\, \boldsymbol{\Gamma}_3 + \sin 4\theta\, \boldsymbol{\Gamma}_4, \tag{2}$$

$$\boldsymbol{\Gamma}_0 = \begin{bmatrix} U_1 & U_4 & 0 \\ U_4 & U_1 & 0 \\ 0 & 0 & U_5 \end{bmatrix}, \quad \boldsymbol{\Gamma}_1 = \begin{bmatrix} U_2 & 0 & 0 \\ 0 & -U_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \boldsymbol{\Gamma}_2 = \tfrac{U_2}{2} \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix},$$
$$\boldsymbol{\Gamma}_3 = U_3 \begin{bmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}, \quad \boldsymbol{\Gamma}_4 = U_3 \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & -1 \\ 1 & -1 & 0 \end{bmatrix},$$

with the material invariants $U_1 = \frac{1}{8}(3Q_{11} + 3Q_{22} + 2Q_{12} + 4Q_{66})$, $U_2 = \frac{1}{2}(Q_{11} - Q_{22})$, $U_3 = \frac{1}{8}(Q_{11} + Q_{22} - 2Q_{12} - 4Q_{66})$, $U_4 = \frac{1}{8}(Q_{11} + Q_{22} + 6Q_{12} - 4Q_{66})$, $U_5 = \frac{1}{8}(Q_{11} + Q_{22} - 2Q_{12} + 4Q_{66})$. $\boldsymbol{\Gamma}_0$ is isotropic. All the dependence on orientation is carried by U_2 and U_3, which vanish for an isotropic material. For the CFRP above, $U_2/U_1 = 1.10$ and $U_3/U_1 = 0.26$.

A symmetric laminate ($\boldsymbol{B} = \boldsymbol{0}$) of thickness h has the membrane and bending stiffnesses

$$\boldsymbol{A} = h\Big[\boldsymbol{\Gamma}_0 + \textstyle\sum_{i=1}^{4} \xi_i^A \boldsymbol{\Gamma}_i\Big], \qquad \boldsymbol{D} = \tfrac{h^3}{12}\Big[\boldsymbol{\Gamma}_0 + \textstyle\sum_{i=1}^{4} \xi_i^D \boldsymbol{\Gamma}_i\Big], \tag{3}$$

with the lamination parameters [21, 25]

$$\boldsymbol{\xi}^A = \int_{-1/2}^{1/2} \boldsymbol{v}(\theta(\bar{z}))\, \mathrm{d}\bar{z}, \qquad \boldsymbol{\xi}^D = 12 \int_{-1/2}^{1/2} \bar{z}^2\, \boldsymbol{v}(\theta(\bar{z}))\, \mathrm{d}\bar{z}, \qquad \boldsymbol{v}(\theta) = (\cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta), \tag{4}$$

where $\theta(\bar{z})$ is the ply angle at normalized height $\bar{z} = z/h$. Two facts organize everything that follows.

- A *single-ply-like* laminate (one angle through the thickness) has $\boldsymbol{\xi}^A = \boldsymbol{\xi}^D = \boldsymbol{v}(\theta)$, so $\boldsymbol{D} = \frac{h^2}{12}\boldsymbol{A}$ pointwise.
- A *stacking sequence* makes $\boldsymbol{\xi}^D \neq \boldsymbol{\xi}^A$. Because $\boldsymbol{\xi}^D$ weights the outer plies by $12\bar{z}^2$, the stacking sequence separates bending from membrane stiffness *through the thickness*.

A *steered* laminate makes $\boldsymbol{\xi}^{A,D}$ functions of the in-plane position $\boldsymbol{x}$.

2.2 The laminated von Kármán plate

A rectangular plate $\Omega = [0, a] \times [0, b]$ ($a = 0.20\,\mathrm{m}$, $b = 0.15\,\mathrm{m}$, $h = 2\,\mathrm{mm}$) carries the transverse deflection w and in-plane displacements $\boldsymbol{u} = (u, v)$. The von Kármán membrane strain and curvature are

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_L(\boldsymbol{u}) + \boldsymbol{\varepsilon}_{NL}(w), \quad \boldsymbol{\varepsilon}_L = \begin{bmatrix} u_{,x} \\ v_{,y} \\ u_{,y} + v_{,x} \end{bmatrix}, \quad \boldsymbol{\varepsilon}_{NL} = \tfrac{1}{2} \begin{bmatrix} w_{,x}^2 \\ w_{,y}^2 \\ 2w_{,x}w_{,y} \end{bmatrix}, \qquad \boldsymbol{\kappa} = -\begin{bmatrix} w_{,xx} \\ w_{,yy} \\ 2w_{,xy} \end{bmatrix}, \tag{5}$$

and, with $\boldsymbol{A}(\boldsymbol{x}), \boldsymbol{D}(\boldsymbol{x})$ from (3), the strain energy is

$$U = \tfrac{1}{2}\int_\Omega \boldsymbol{\kappa}^\top \boldsymbol{D}\, \boldsymbol{\kappa}\, \mathrm{d}\Omega + \tfrac{1}{2}\int_\Omega (\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL})^\top \boldsymbol{A}\, (\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL})\, \mathrm{d}\Omega. \tag{6}$$

With the kinetic energy $T = \frac{1}{2}\int \rho h(\dot{w}^2 + |\dot{\boldsymbol{u}}|^2)$, Hamilton's principle gives the laminated Föppl–von Kármán equations [11],

$$N_{\alpha\beta,\beta} = \rho h\, \ddot{u}_\alpha, \qquad M_{\alpha\beta,\alpha\beta} + \big(N_{\alpha\beta} w_{,\beta}\big)_{,\alpha} + p = \rho h\, \ddot{w}, \qquad \boldsymbol{N} = \boldsymbol{A}(\boldsymbol{\varepsilon}_L + \boldsymbol{\varepsilon}_{NL}),\ \boldsymbol{M} = \boldsymbol{D}\boldsymbol{\kappa}. \tag{7}$$

The edges are simply supported and immovable ($w = 0$, $\boldsymbol{u} = \boldsymbol{0}$ on $\partial\Omega$). We discretize by Galerkin's method on sine bases, which satisfy these conditions term by term:

$$w = \sum_{k=1}^{N_w} q_k\, \phi_k, \quad \phi_k = \sin\tfrac{m_k \pi x}{a} \sin\tfrac{n_k \pi y}{b}; \qquad u, v = \sum_{j=1}^{N_s} s_j^{u,v}\, \psi_j, \quad \psi_j = \sin\tfrac{i_j \pi x}{a} \sin\tfrac{l_j \pi y}{b}. \tag{8}$$

We use $N_w = 64$ ($m, n \le 8$) and $N_s = 36$ per component. The bases are converged for the steered designs: Table 1 shows that the steered relaxation needs at least five to six in-plane modes per direction; the three-mode basis that suffices for a uniform laminate under-resolves it badly ($J/J_{\mathrm{UD}} = 0.52$ instead of 0.37). With $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{s}]$ the system is

$$\boldsymbol{M}\ddot{\boldsymbol{x}} + \boldsymbol{C}\dot{\boldsymbol{x}} + \boldsymbol{K}\boldsymbol{x} + \boldsymbol{B}_2(\boldsymbol{x}, \boldsymbol{x}) + \boldsymbol{T}_3(\boldsymbol{x}, \boldsymbol{x}, \boldsymbol{x}) = \boldsymbol{f}(t), \tag{9}$$

with $M_{kl} = \rho h \int \phi_k \phi_l$ (likewise for ψ), $\boldsymbol{K} = \mathrm{blkdiag}(\boldsymbol{K}_b, \boldsymbol{K}_{\mathrm{in}})$, $(\boldsymbol{K}_b)_{kl} = \int \boldsymbol{\kappa}_k^\top \boldsymbol{D}\boldsymbol{\kappa}_l$ and $(\boldsymbol{K}_{\mathrm{in}})_{jl} = \int \boldsymbol{\varepsilon}_{L,j}^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{L,l}$. $\boldsymbol{B}_2$ is the gradient of $\int \boldsymbol{\varepsilon}_L^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$ (the $\boldsymbol{u} \leftrightarrow w^2$ coupling) and $\boldsymbol{T}_3$ is the gradient of $\frac{1}{2}\int \boldsymbol{\varepsilon}_{NL}^\top \boldsymbol{A}\boldsymbol{\varepsilon}_{NL}$ (the membrane cubic). Because $\boldsymbol{B} = \boldsymbol{0}$, the *linear* problem decouples into a bending block and an in-plane block. The mass is design-independent: every design in this paper has the same plate, the same material and the same thickness.

The *beam* is the one-dimensional specialization: a clamped–clamped strip ($L = 0.30\,\mathrm{m}$, width $20\,\mathrm{mm}$, $h = 2\,\mathrm{mm}$) with axial u and transverse w, $\varepsilon = u' + \frac{1}{2}w'^2$ and $\kappa = -w''$. For a single-ply laminate, $A = bhE(x)$ and $D = \frac{bh^3}{12}E(x)$ with $E(x) = \bar{Q}_{11}(\theta(x))$. It is discretized by Hermite-cubic/linear elements, and the axial ends are immovable.

2.3 Spectral submanifold and backbone

In first-order form $\boldsymbol{z} = [\boldsymbol{x}; \dot{\boldsymbol{x}}]$, $\boldsymbol{\mathcal{B}}\dot{\boldsymbol{z}} = \boldsymbol{\mathcal{A}}\boldsymbol{z} + \boldsymbol{F}(\boldsymbol{z})$ with $\boldsymbol{\mathcal{B}} = \left[\begin{smallmatrix} \boldsymbol{C} & \boldsymbol{M} \\ \boldsymbol{M} & \boldsymbol{0} \end{smallmatrix}\right]$, $\boldsymbol{\mathcal{A}} = \left[\begin{smallmatrix} -\boldsymbol{K} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{M} \end{smallmatrix}\right]$ and $\boldsymbol{F} = [-\boldsymbol{B}_2 - \boldsymbol{T}_3; \boldsymbol{0}]$. Let $\lambda_1, \bar{\lambda}_1$ be the eigenvalues of the fundamental, with right eigenvector $\boldsymbol{v}_1$ and left eigenvector $\boldsymbol{u}_1$ ($\boldsymbol{u}_1^* \boldsymbol{\mathcal{A}} = \lambda_1 \boldsymbol{u}_1^* \boldsymbol{\mathcal{B}}$, $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}} \boldsymbol{v}_1 = 1$). The SSM $\mathcal{W}$ is the unique smoothest invariant manifold tangent to their eigenspace [3]. It exists if λ_1 is the slowest pair and no outer eigenvalue equals $m_1\lambda_1 + m_2\bar{\lambda}_1$ up to the truncation order. Parametrizing $\boldsymbol{z} = \boldsymbol{W}(\boldsymbol{p})$ and $\dot{\boldsymbol{p}} = \boldsymbol{R}(\boldsymbol{p})$ gives the invariance equation $\boldsymbol{\mathcal{B}}\,\mathrm{D}\boldsymbol{W}\,\boldsymbol{R} = \boldsymbol{\mathcal{A}}\boldsymbol{W} + \boldsymbol{F}(\boldsymbol{W})$. Expanding both maps in monomials $\boldsymbol{p}^{\boldsymbol{m}}$, it separates order by order into the homological equations

$$(\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})\boldsymbol{w}_{\boldsymbol{m}} + \boldsymbol{\mathcal{B}}(\boldsymbol{v}_1 R^1_{\boldsymbol{m}} + \bar{\boldsymbol{v}}_1 R^2_{\boldsymbol{m}}) = \boldsymbol{G}_{\boldsymbol{m}}, \qquad \Lambda_{\boldsymbol{m}} = m_1\lambda_1 + m_2\bar{\lambda}_1, \tag{10}$$

where $\boldsymbol{G}_{\boldsymbol{m}}$ collects lower-order terms (Appendix A). At the near-resonant indices $\boldsymbol{m} = (k{+}1, k)$ the operator is (near-)singular. Projecting onto $\boldsymbol{u}_1$ gives the reduced (normal-form) coefficients $\gamma_{2k+1} = \boldsymbol{u}_1^* \boldsymbol{G}_{(k+1,k)}$. On the orbit $p_1 = \frac{\rho}{2}e^{i\vartheta}$, $p_2 = \bar{p}_1$, the backbone and the modal decay rate are

$$\Omega(\rho) = \mathrm{Im}\left(\lambda_1 + \sum_{k\ge 1} \gamma_{2k+1}\big(\tfrac{\rho}{2}\big)^{2k}\right), \qquad \alpha(\rho) = \mathrm{Re}(\cdots). \tag{11}$$

The implementation is verified against the analytic Duffing backbone through fifth order ($3/8$ and $-21/256$), against full-order EPMC continuation, and against two published results [15] (Appendix D). The master eigenvector is computed by Rayleigh-quotient inverse iteration on the complex-symmetric pencil, with $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$. A dense QZ solve of the stiff first-order pencil was found to lose up to ten digits (eigenvector condition number $\sim 10^{14}$ for laminated beams) and is no longer used for symmetric systems (regression test in `tests/test_relaxed_plate.py`).

2.4 The relaxed hardening coefficient

Let the fundamental be the lowest Ritz pair of the bending block, $\boldsymbol{K}_b\boldsymbol{v} = \omega_1^2\boldsymbol{M}_w\boldsymbol{v}$ with $\boldsymbol{v}^\top\boldsymbol{M}_w\boldsymbol{v} = 1$. Its shape $w_1 = \sum_k v_k\phi_k$ *adapts* to the stiffness field. This is the exact linear master mode of (9), because the blocks decouple when $\boldsymbol{B} = \boldsymbol{0}$. Write its curvature as $\boldsymbol{\kappa}_1$ and its stretch as

$$\boldsymbol{m} = \left[w_{1,x}^2,\ w_{1,y}^2,\ 2w_{1,x}w_{1,y}\right]^\top = 2\boldsymbol{\varepsilon}_{NL}(w_1). \tag{12}$$

Proposition 1 (Relaxed cubic, variational form)**.** *Let the in-plane field be slaved quasi-statically to q^2, the square of the fundamental's modal amplitude. Then the cubic coefficient of the single-mode reduced dynamics $\ddot{q} + \omega_1^2 q + \gamma q^3 = 0$ is*

$$\gamma = \min_{\boldsymbol{s}}\ \tfrac{1}{2}\int_\Omega (\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))^\top \boldsymbol{A}\,(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))\,\mathrm{d}\Omega = k_{nl} - 2\,\boldsymbol{g}^\top\boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g} = \tfrac{1}{2}\int_\Omega \boldsymbol{\varepsilon}_r^\top\boldsymbol{A}\,\boldsymbol{\varepsilon}_r\,\mathrm{d}\Omega, \tag{13}$$

with $k_{nl} = \frac{1}{2}\int \boldsymbol{m}^\top\boldsymbol{A}\boldsymbol{m}$, $\boldsymbol{g} = \frac{1}{2}\int \boldsymbol{\varepsilon}_L(\cdot)^\top\boldsymbol{A}\boldsymbol{m}$, $\boldsymbol{s}^\star = \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$ and the relaxed stretch $\boldsymbol{\varepsilon}_r = \boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}^\star)$.

Proof. With $w = qw_1$ and $\boldsymbol{u} = -q^2\boldsymbol{s}$, the membrane strain is $\boldsymbol{\varepsilon} = \frac{q^2}{2}(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))$. The membrane energy is therefore $\frac{q^4}{8}\int(\boldsymbol{m} - 2\boldsymbol{\varepsilon}_L(\boldsymbol{s}))^\top\boldsymbol{A}(\cdot) = \frac{q^4}{4}F(\boldsymbol{s})$, where $F(\boldsymbol{s}) = k_{nl} - 4\boldsymbol{s}^\top\boldsymbol{g} + 2\boldsymbol{s}^\top\boldsymbol{K}_{\mathrm{in}}\boldsymbol{s}$. Static equilibrium of the in-plane field minimizes F, at $\boldsymbol{s}^\star = \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$, with $F(\boldsymbol{s}^\star) = k_{nl} - 2\boldsymbol{g}^\top\boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$. The reduced potential $\frac{\gamma}{4}q^4$ then gives (13). The last form is $F(\boldsymbol{s}^\star)$ written out. □

Proposition 1 has three uses. (i) k_{nl} is the restrained (immovable in-plane field) upper bound, and $2\boldsymbol{g}^\top\boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$ is the *relaxation*. Its share of k_{nl} is 33% for the unidirectional plate. (ii) Because γ is a *minimum* over $\boldsymbol{s}$, its design derivative needs no in-plane adjoint (envelope theorem; Sec. 4.3). (iii) It writes the hardening as the integral of an energy density (Sec. 3).

Proposition 2 (Agreement with the SSM)**.** *For an undamped system* (9) *with $\boldsymbol{B} = \boldsymbol{0}$, the cubic SSM coefficient of the fundamental satisfies* $\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}\gamma\,[1 + O(4\omega_1^2/\omega_{\mathrm{in}}^2)]$*, where ω_{in} is the lowest in-plane frequency.*

The proof is in Appendix B. In short: the $(1,1)$ in-plane homological solve is exactly static ($\Lambda_{(1,1)} = 0$), and the $(2,0)$ solve is static up to the resolvent factor $(\boldsymbol{K}_{\mathrm{in}} - 4\omega_1^2\boldsymbol{M}_{\mathrm{in}})^{-1}$. For our plate $4\omega_1^2/\omega_{\mathrm{in}}^2 = 3.2\times10^{-3}$ ($f_{\mathrm{in}} \approx 9.0\,\mathrm{kHz}$ on the full multi-mode plate), and (13) reproduces the full multi-mode SSM coefficient to within 1.2% (Sec. 5.5).

2.5 The backbone in physical amplitude, and the design quantity J

For $\ddot{q} + \omega_1^2 q + \gamma q^3 = 0$ the cubic SSM gives $\Omega = \omega_1 + \frac{3}{8}(\gamma/\omega_1)q^2$. The mid-plate deflection of the mode is $a = w_c q$, with $w_c = w_1(a/2, b/2)$, so

$$\frac{\Omega}{\omega_1} = 1 + \frac{3}{8}J\,a^2 = 1 + \beta\Big(\frac{a}{h}\Big)^2, \qquad J = \frac{\gamma}{\omega_1^2 w_c^2}, \qquad \beta = \tfrac{3}{8}Jh^2. \tag{14}$$

J (units m^{-2}) is the *frequency-normalized detuning*. It is invariant to the mode normalization, and it is what we design. The unidirectional plate has $f_1 = 256.3\,\mathrm{Hz}$, $\beta = 0.684$ and a 10.95% shift at $a/h = 0.4$. The cubic truncation reproduces the full-order EPMC detuning to within 5% over the amplitudes EPMC reached, $a/h \le 0.21$ (UD) and ≤ 0.35 (steered) (Appendix D).

2.6 Forced response and the jump-free amplitude

With harmonic forcing $\boldsymbol{f}\cos\Omega t$ and damping $\boldsymbol{C} = 2\zeta\omega_1\boldsymbol{M}$, seeking $p_1 = \frac{\rho}{2}e^{i\Omega t}$ on the SSM gives the leading-order forced-response curve

$$\big[(\Omega - \Omega(\rho))^2 + \alpha(\rho)^2\big]\rho^2 = |f_1|^2, \qquad f_1 = \boldsymbol{u}_1^*[\boldsymbol{f};\boldsymbol{0}]. \tag{15}$$

Proposition 3 (Jump-free amplitude)**.** *For the cubic backbone* (14) *with* $\alpha = \zeta\omega_1$*, the forced response* (15) *is single-valued in* Ω *for every drive whose peak (resonant) amplitude is below*

$$\frac{a_c}{h} = \Big(\frac{8\zeta}{3\sqrt{3}\,\beta}\Big)^{1/2} \;\propto\; J^{-1/2}, \tag{16}$$

and multivalued (a fold pair with jump and hysteresis) above it. At the critical drive the vertical tangent sits at the lower amplitude $(\sqrt{3}/2)\,a_c$ *on the response curve.*

Proof. Write $x = a^2$, $c = \omega_1\beta/h^2$, $\mu = \zeta\omega_1$, and let F be the drive expressed in physical amplitude. Then (15) reads $[(\sigma - cx)^2 + \mu^2]x = F^2$, with $\sigma = \Omega - \omega_1$. Put $\sigma = cx + s$. The curve is $x = F^2/(s^2+\mu^2)$, $\sigma = cF^2/(s^2+\mu^2)+s$, and it folds where $\mathrm{d}\sigma/\mathrm{d}s = 0$, i.e. where $2cF^2s = (s^2+\mu^2)^2$. A fold pair first appears when this equation has a double root, i.e. when $2cF^2 = 4s(s^2+\mu^2)$ also holds. Combining the two conditions gives $s^2 + \mu^2 = 4s^2$, so $s = \mu/\sqrt{3}$, and the critical drive is $F_c^2 = (s^2+\mu^2)^2/(2cs) = 8\mu^3/(3\sqrt{3}\,c)$. The vertical tangent lies at $x = F_c^2/(s^2+\mu^2) = 2\mu/(\sqrt{3}c)$, while the peak of the response (at $s = 0$) is $x_c = F_c^2/\mu^2 = 8\mu/(3\sqrt{3}\,c)$, which is (16). Below F_c the curve has no fold; above it, it has two. □

Proposition 3 is the consequence we test. Holding ω_1, ζ and h and reducing J by a factor r widens the jump-free amplitude by $r^{-1/2}$. In Sec. 5.5 we compute a_c *numerically*, as the drive at which the full-order damped SSM response first becomes multivalued, and compare it with (16).

3 Theory: why a fiber field can decouple hardening from frequency

3.1 Two energy densities over one stiffness

With the mass-normalized adapted mode, the Rayleigh quotient and Proposition 1 write both design quantities as integrals of local energy densities:

$$\omega_1^2 = \int_\Omega e_K\,\mathrm{d}\Omega, \quad e_K = \boldsymbol{\kappa}_1^\top\boldsymbol{D}\,\boldsymbol{\kappa}_1; \qquad \gamma = \int_\Omega e_G\,\mathrm{d}\Omega, \quad e_G = \tfrac{1}{2}\boldsymbol{\varepsilon}_r^\top\boldsymbol{A}\,\boldsymbol{\varepsilon}_r. \tag{17}$$

$J = \gamma/(\omega_1^2 w_c^2)$ is their ratio up to the mode's mid-plate value. Both densities read the *same* rotated material (2), but through different strain fields: $\boldsymbol{\kappa}_1$ (second derivatives of w_1, largest where the mode bends) and $\boldsymbol{\varepsilon}_r$ (squared first derivatives minus the in-plane relief). For the simply supported fundamental both peak at the center, but with different principal directions (Sec. 5.3). The following proposition shows exactly how each density depends on the local fiber angle.

3.2 Local Fourier structure

Proposition 4 (Angular dependence of a laminate energy density)**.** *Let* $\boldsymbol{s}$ *be an engineering strain vector whose tensor has principal values* $e_{\max}, e_{\min}$*, ordered so that* $|e_{\max}| \geq |e_{\min}|$*, and let* $\hat{\varphi}$ *be the*

direction of $e_{\max}$. *Then for the single-ply stiffness* (2),

$$\boldsymbol{s}^\top \bar{\boldsymbol{Q}}(\theta)\boldsymbol{s} = \underbrace{U_1(e_{\max}^2 + e_{\min}^2) + 2U_4 e_{\max} e_{\min}}_{isotropic} + U_2\,\Delta\cos 2(\theta-\hat{\varphi}) + U_3\,\Sigma\cos 4(\theta-\hat{\varphi}), \tag{18}$$

with $\Delta = e_{\max}^2 - e_{\min}^2 \geq 0$ *and* $\Sigma = (e_{\max}-e_{\min})^2 \geq 0$.

Proof. Rotation invariance gives $\boldsymbol{s}^\top\bar{\boldsymbol{Q}}(\theta)\boldsymbol{s} = \boldsymbol{s}'^\top\bar{\boldsymbol{Q}}(\theta-\hat{\varphi})\boldsymbol{s}'$, with $\boldsymbol{s}' = [e_{\max}, e_{\min}, 0]^\top$ the strain in its principal frame. Let $\psi = \theta - \hat{\varphi}$. Using (2), $\bar{Q}_{11} = U_1 + U_2\cos 2\psi + U_3\cos 4\psi$, $\bar{Q}_{22} = U_1 - U_2\cos 2\psi + U_3\cos 4\psi$ and $\bar{Q}_{12} = U_4 - U_3\cos 4\psi$, we get $\bar{Q}_{11}e_{\max}^2 + 2\bar{Q}_{12}e_{\max}e_{\min} + \bar{Q}_{22}e_{\min}^2 = U_1(e_{\max}^2+e_{\min}^2) + 2U_4e_{\max}e_{\min} + U_2(e_{\max}^2 - e_{\min}^2)\cos 2\psi + U_3(e_{\max}-e_{\min})^2\cos 4\psi$. □

Applying (18) to the two densities, pointwise:

$$e_K(\boldsymbol{x},\theta) = \tfrac{h^3}{12}\big[I_K + U_2\Delta_K\cos 2(\theta-\varphi_K) + U_3\Sigma_K\cos 4(\theta-\varphi_K)\big], \tag{19}$$

$$e_G(\boldsymbol{x},\theta) = \tfrac{h}{2}\big[I_G + U_2\Delta_G\cos 2(\theta-\varphi_G) + U_3\Sigma_G\cos 4(\theta-\varphi_G)\big], \tag{20}$$

where $(\Delta_K, \Sigma_K, \varphi_K)$ come from the principal curvatures of w_1 and $(\Delta_G, \Sigma_G, \varphi_G)$ from the principal relaxed stretches. For CFRP $U_2 \approx 4.2\,U_3$, so the 2θ harmonic dominates. To leading order, then:

- e_K is largest (the point contributes most to ω_1^2) when the fiber lies *along* the principal curvature, $\theta = \varphi_K$;
- e_G is smallest (the point contributes least to γ) when the fiber lies *across* the principal stretch, $\theta = \varphi_G + \pi/2$.

Which way the frequency pulls depends on the sign of the multiplier Λ of Proposition 5 below, which weights e_K against e_G. If the target frequency lies above the one the stretch alone would select, $\Lambda > 0$ and the fiber is drawn toward the principal curvature, $\theta = \varphi_K$. If it lies below, $\Lambda < 0$ and the fiber is pushed *across* the principal curvature, $\theta = \varphi_K + \pi/2$, the bending-compliant direction. We call $\varphi_G + \pi/2$ and the frequency-favored direction the *preferred directions*, and the angle between them, $\chi(\boldsymbol{x}) \in [0, \pi/2]$, the *conflict*. Where $\chi = 0$, one fiber direction serves both energies. Where χ is large, any single direction compromises.

Corollary 1 (The unidirectional floor)**.** *Among all uniform symmetric laminates of the same thickness and mass, whatever the stacking sequence, the lowest linear frequency is attained by a single-angle laminate.*

Proof. For a symmetric laminate $\boldsymbol{D} = \frac{h^3}{12}\big(\boldsymbol{\Gamma}_0 + \sum_k \xi_k^D\boldsymbol{\Gamma}_k\big)$ is affine in its four lamination parameters $\boldsymbol{\xi}^D$, and the set of $\boldsymbol{\xi}^D$ reachable by stacking lies in the convex hull of the single-angle curve $\boldsymbol{\xi}^D(\theta) = (\cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta)$ [21]. ω_1^2 is the minimum over admissible w of a Rayleigh quotient that is affine in $\boldsymbol{\xi}^D$, hence a concave function of $\boldsymbol{\xi}^D$, and a concave function on a convex hull attains its minimum at an extreme point, i.e. at a single-angle laminate. □

For the plate studied here the single-angle frequency rises monotonically from $\theta = 0°$ (Sec. 5.2), so UD is the lowest-frequency uniform laminate of *any* stacking sequence, not only of the enumerated ones, and at $\omega_{1,\mathrm{UD}}$ there is no other uniform competitor.

Corollary 2 (P1: isotropy annihilates the lever)**.** *If* $U_2 = U_3 = 0$*, then* $\boldsymbol{A}$ *and* $\boldsymbol{D}$ *are independent of every ply angle at every point. Hence* ω_1*,* γ *and* J *are independent of the orientation field, uniform or steered or stacked, and every orientation sensitivity vanishes identically. This is exact, including mode adaptation and relaxation, because the operators themselves do not depend on* θ*.*

Numerically, $\partial J/\partial\theta \equiv 0$ to 10^{-10} relative when $\boldsymbol{Q}$ is set isotropic (`tests/test_relaxed_plate.py`).

3.3 Two separation channels

Inserting (3) into (17) gives

$$e_K = \tfrac{h^3}{12}\Big[\boldsymbol{\kappa}_1^\top \boldsymbol{\Gamma}_0 \boldsymbol{\kappa}_1 + \sum_i \xi_i^D(\boldsymbol{x})\, \boldsymbol{\kappa}_1^\top \boldsymbol{\Gamma}_i \boldsymbol{\kappa}_1\Big], \qquad e_G = \tfrac{h}{2}\Big[\boldsymbol{\varepsilon}_r^\top \boldsymbol{\Gamma}_0 \boldsymbol{\varepsilon}_r + \sum_i \xi_i^A(\boldsymbol{x})\, \boldsymbol{\varepsilon}_r^\top \boldsymbol{\Gamma}_i \boldsymbol{\varepsilon}_r\Big]. \tag{21}$$

The frequency reads only the bending lamination parameters, and the hardening only the membrane ones. Each design class corresponds to a different freedom in the field $(\boldsymbol{\xi}^A(\boldsymbol{x}), \boldsymbol{\xi}^D(\boldsymbol{x}))$:

class	$\boldsymbol{\xi}^A$	$\boldsymbol{\xi}^D$	separation available
uniform single angle	$\boldsymbol{v}(\theta_0)$	$= \boldsymbol{\xi}^A$	none (one scalar)
uniform stack	const.	const. $\neq \boldsymbol{\xi}^A$	through the thickness
steered single ply	$\boldsymbol{v}(\theta(\boldsymbol{x}))$	$= \boldsymbol{\xi}^A$	in the plane
steered stack	$\mathcal{R}_{\theta(\boldsymbol{x})}\boldsymbol{\xi}_0^A$	$\mathcal{R}_{\theta(\boldsymbol{x})}\boldsymbol{\xi}_0^D$	both

Here $\mathcal{R}_\theta$ rotates the parameter pairs, $(\xi_1 + i\xi_2) \mapsto e^{2i\theta}(\xi_1 + i\xi_2)$ and $(\xi_3 + i\xi_4) \mapsto e^{4i\theta}(\xi_3 + i\xi_4)$. Section 5.2 measures the achievable (ω_1, J) set of each class.

3.4 Pointwise optimality: what an optimal field must do

Consider the problem

$$\min_{\theta(\cdot)} \; J \quad \text{s.t.} \quad \omega_1 = \omega_\star \tag{22}$$

over single-ply fields. Freeze w_1 and $\boldsymbol{\varepsilon}_r$ at a reference design. This *frozen-field* model is exact to first order in the design change: $\boldsymbol{s}^\star$ is stationary (Proposition 1) and ω_1^2 is a Rayleigh quotient. In this model w_c is fixed, so (22) reads $\min \int e_G(\boldsymbol{x}, \theta(\boldsymbol{x}))$ subject to $\int e_K(\boldsymbol{x}, \theta(\boldsymbol{x})) = \omega_\star^2$.

Proposition 5 (Pointwise optimality and strict benefit of steering)**.** *In the frozen-field model the optimal field is*

$$\theta^\star(\boldsymbol{x}) = \arg\min_\theta \big[e_G(\boldsymbol{x}, \theta) - \Lambda\, e_K(\boldsymbol{x}, \theta)\big], \tag{23}$$

with the scalar multiplier Λ fixed by the frequency constraint. A uniform angle θ_u can be optimal among all fields only if

$$U_2\Delta_G \sin 2(\theta_u - \varphi_G) + 2U_3\Sigma_G \sin 4(\theta_u - \varphi_G) = \Lambda'\big[U_2\Delta_K \sin 2(\theta_u - \varphi_K) + 2U_3\Sigma_K \sin 4(\theta_u - \varphi_K)\big] \tag{24}$$

holds at almost every $\boldsymbol{x}$, with one constant $\Lambda' = \frac{h^2}{6}\Lambda$. Otherwise a steered field strictly lowers J at the same frequency.

Proof. The Lagrangian $\int e_G - \Lambda(\int e_K - \omega_\star^2)$ has no derivative coupling between points, so it is minimized pointwise, giving (23). Every uniform angle is a feasible field, so the field optimum is never worse. If the uniform optimum were also the field optimum, it would satisfy the pointwise stationarity condition $\partial_\theta e_G = \Lambda \partial_\theta e_K$ at every point. By (19)–(20) this is (24). □

Condition (24) asks the anisotropic parts of two different strain fields to be proportional with one constant at a single angle. That fails whenever the conflict $\chi(\boldsymbol{x})$ or the ratio Δ_G/Δ_K varies over the plate. Since the principal curvature and the principal relaxed stretch of a bending mode turn differently over the plate, and their energy ratio varies, it fails generically. Proposition 5 is the precise form of the hypothesis. It also yields a test (P4): $\theta^\star$ computed from the *unidirectional* fields alone, with no optimizer, should already capture much of the gain, and the optimized field should resemble it.

3.5 One dimension in closed form

Proposition 6 (Beam: harmonic against curvature-weighted mean)**.** *Consider a symmetric single-ply beam with immovable ends, $A = bhE(x)$ and $D = \frac{bh^3}{12}E(x)$, and a mass-normalized fundamental φ with mid-span value φ_c. Then*

$$\gamma = \frac{bh\,(\int_0^L \varphi'^2)^2}{2\int_0^L E^{-1}}, \qquad \omega_1^2 = \frac{bh^3}{12}\int_0^L E\,\varphi''^2, \qquad J = \frac{6}{h^2\varphi_c^2}\,\frac{(\int \varphi'^2)^2}{\int E^{-1}\int E\varphi''^2}. \tag{25}$$

Proof. Axial equilibrium makes N uniform. Integrating $\varepsilon = N/A = u' + \frac{1}{2}q^2\varphi'^2$ over the span with $u(0) = u(L) = 0$ gives $N\int A^{-1} = \frac{1}{2}q^2\int \varphi'^2$. The membrane energy is then $\frac{1}{2}N^2\int A^{-1} = q^4(\int \varphi'^2)^2/(8\int A^{-1})$, which equals $\frac{\gamma}{4}q^4$. The frequency is the Rayleigh quotient, and J follows from (14). □

Corollary 3 (P2: uniform invariance)**.** *For uniform $E \equiv E_0$, $\int E^{-1}\int E\varphi''^2 = L\int \varphi''^2$, and φ itself is independent of E_0 (the operators scale together). Hence J is* exactly *independent of any uniform single-ply angle, and every change in J is caused by spatial variation of $E(x)$.*

Equation (25) makes the mechanism transparent. The hardening sees the *harmonic* mean of E over the whole span, which is dominated by the softest segment. The frequency sees the mean of E *weighted by the curvature* φ''^2, which vanishes at the inflection points ($x/L \approx 0.224$ and 0.776 for the clamped fundamental). Compliant fibers placed there raise $\int E^{-1}$, and so lower the hardening, at almost no cost in frequency.

Corollary 4 (Bang-bang bound)**.** *Suppose $E = E_{\max}$ except on zones of total length ℓ where $E = E_{\min}$ and $\varphi'' \approx 0$. Then, to frozen-mode accuracy,*

$$J/J_{\mathrm{UD}} \approx \left[1 + (\ell/L)(E_{\max}/E_{\min} - 1)\right]^{-1}. \tag{26}$$

For CFRP, $E_{\max}/E_{\min} = \bar{Q}_{11}(0)/\bar{Q}_{11}(90^\circ) = 14.6$, so $\ell/L = 0.05$ already gives $J/J_{\mathrm{UD}} \approx 0.59$. Equation (26) contains P5 in one dimension. The lever grows with the anisotropy ratio and with how narrow the compliant zones can be made while staying where $\varphi'' \approx 0$. The narrowness is set by the tow curvature: turning a tow by 90° at rate κ_{tow} needs a run of $(\pi/2)/\kappa_{\mathrm{tow}}$, which is 0.10 m at 900°/m, a third of the span.

4 Design problems and sensitivities

4.1 Orientation field and the tow-curvature constraint

The fiber angle is a smooth field interpolated from M support values $\hat{\boldsymbol{\theta}}$ by normalized compactly supported Wendland C^2 radial basis functions [46],

$$\theta(\boldsymbol{x}) = \sum_{i=1}^{M} \Phi_i(\boldsymbol{x})\hat{\theta}_i, \qquad \Phi_i = \frac{\phi(\|\boldsymbol{x} - \boldsymbol{x}_i\|/r_s)}{\sum_j \phi(\|\boldsymbol{x} - \boldsymbol{x}_j\|/r_s)}, \qquad \phi(q) = (1-q)_+^4(4q+1), \tag{27}$$

on a regular support grid of spacing h_s with $r_s = 1.5h_s$. The baseline is $h_s = 50$ mm ($M = 20$); Sec. 5.4 refines it. The spatial derivatives $\boldsymbol{\Phi}_x, \boldsymbol{\Phi}_y$ are analytic. A tow laid along $\boldsymbol{n} = (\cos\theta, \sin\theta)$ turns at the rate

$$\kappa_{\mathrm{tow}}(\boldsymbol{x}) = (\boldsymbol{n}\cdot\nabla)\theta = \cos\theta\,\theta_{,x} + \sin\theta\,\theta_{,y}, \tag{28}$$

so its steering radius is $1/|\kappa_{\text{tow}}|$. AFP and continuous-fiber printing require $|\kappa_{\text{tow}}| \le \zeta_{\text{all}} = 1/R_{\min}$; this pointwise form follows [44]. We impose this at every quadrature point e as $c_e = 1-\kappa_{\text{tow},e}^2/\zeta_{\text{all}}^2 \ge 0$, with the analytic Jacobian

$$\frac{\partial \kappa_{\text{tow},e}}{\partial \hat{\boldsymbol{\theta}}} = \big(-\sin\theta_e\, \boldsymbol{\Phi}_{x,e}\hat{\boldsymbol{\theta}} + \cos\theta_e\, \boldsymbol{\Phi}_{y,e}\hat{\boldsymbol{\theta}}\big)\boldsymbol{\Phi}_e + \cos\theta_e\, \boldsymbol{\Phi}_{x,e} + \sin\theta_e\, \boldsymbol{\Phi}_{y,e}. \tag{29}$$

The baseline limit is $R_{\min} = 20\,\text{mm}$.

4.2 The design problems

Steered plate. For a target frequency $\omega_\star$,

$$\min_{\hat{\boldsymbol{\theta}}} \ \frac{J(\hat{\boldsymbol{\theta}})}{J_{\text{UD}}} \quad \text{s.t.} \quad \frac{\omega_1(\hat{\boldsymbol{\theta}})}{\omega_\star} - 1 = 0, \qquad c_e(\hat{\boldsymbol{\theta}}) \ge 0 \ \ \forall e, \tag{30}$$

where J and ω_1 are those of the adapted-mode, relaxed model (Sec. 2.4). The frequency is an *equality*, so a gain can never come from a stiffer or softer panel. The mass is fixed by construction. We solve (30) by sequential quadratic programming (SLSQP [45]), which satisfies both constraints to 10^{-4}, from six starts (uniform $0°, 20°, 45°, 70°$ and two random fields), and keep the best feasible result.

Uniform stacks. We enumerate every symmetric eight-ply laminate $[\theta_4/\theta_3/\theta_2/\theta_1]_s$ with $\theta_i \in \{-75°, -60°, \dots, 90°\}$: $12^4 = 20{,}736$ stacks, each evaluated exactly on the same model. A uniform laminate needs no optimizer, because its operators are linear combinations of precomputed moment tensors (e.g. $\boldsymbol{K}_b = \sum_{ab} D_{ab}\boldsymbol{I}^K_{ab}$).

Steered stack. The best uniform stack at $\omega_\star$ is rotated pointwise, $\boldsymbol{C}(\boldsymbol{x}) = \boldsymbol{T}(\theta(\boldsymbol{x}))\boldsymbol{C}_0\boldsymbol{T}(\theta(\boldsymbol{x}))^\top$ for $\boldsymbol{C} \in \{\boldsymbol{A}, \boldsymbol{D}\}$, and (30) is solved for $\theta(\boldsymbol{x})$.

Beam. $\theta(x)$ is mirror symmetric about mid-span and piecewise linear on 13 knots over the half span. The tow limit is the slope bound $|\theta'| \le \kappa_{\text{tow}}$, and the frequency is held as $\omega_1 \ge f_{\text{hold}}\,\omega_{1,\text{UD}}$. A uniform-angle beam cannot change J (Corollary 3) and every steered beam is *softer* than UD, so an equality at $f_{\text{hold}} = 1$ admits only UD. We report $f_{\text{hold}} \in \{0.98, 0.90\}$ and $\kappa_{\text{tow}} \in \{900, 1800, \infty\}$ deg/m. The optimizer works on the condensed beam evaluator (adapted mode, exact axial condensation), and every design is *confirmed* on the full FE beam by its cubic SSM.

4.3 Sensitivities on the adapted mode

Let a perturbation $\delta\theta(\boldsymbol{x})$ change the stiffnesses by $\delta\boldsymbol{D} = \boldsymbol{D}'\delta\theta$ and $\delta\boldsymbol{A} = \boldsymbol{A}'\delta\theta$, with $\boldsymbol{C}' = \boldsymbol{T}'\boldsymbol{C}_0\boldsymbol{T}^\top + \boldsymbol{T}\boldsymbol{C}_0\boldsymbol{T}'^\top$. Let $(\lambda_j, \boldsymbol{v}_j)$ be the Ritz pairs of the bending block ($\lambda_1 = \omega_1^2$), with curvature fields $\boldsymbol{\kappa}_j$. The mass does not depend on θ.

Frequency. The Rayleigh quotient is stationary (Fox–Kapoor), so

$$\delta\omega_1^2 = \int \boldsymbol{\kappa}_1^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1. \tag{31}$$

Mode shape. The eigenvector derivative is

$$\delta\boldsymbol{v}_1 = \sum_{j\ge 2} \frac{\boldsymbol{v}_j^\top \delta\boldsymbol{K}_b \boldsymbol{v}_1}{\lambda_1 - \lambda_j}\boldsymbol{v}_j, \qquad \boldsymbol{v}_j^\top \delta\boldsymbol{K}_b\boldsymbol{v}_1 = \int \boldsymbol{\kappa}_j^\top \delta\boldsymbol{D}\, \boldsymbol{\kappa}_1. \tag{32}$$

Hardening. By Proposition 1, γ is a minimum over the in-plane state, so its total derivative needs *no in-plane adjoint*: only the explicit $\boldsymbol{A}$-dependence and the dependence on the mode (through $\boldsymbol{m}$) remain. With $\partial\gamma/\partial\boldsymbol{m} = \boldsymbol{A}\boldsymbol{\varepsilon}_r$ and $\partial\boldsymbol{m}/\partial v_k = [2w_{1,x}\phi_{k,x},\, 2w_{1,y}\phi_{k,y},\, 2(w_{1,x}\phi_{k,y} + w_{1,y}\phi_{k,x})]$,

$$\delta\gamma = \int \Big[\tfrac{1}{2}\boldsymbol{\varepsilon}_r^\top \delta\boldsymbol{A}\,\boldsymbol{\varepsilon}_r + \boldsymbol{\kappa}_\beta^\top \delta\boldsymbol{D}\,\boldsymbol{\kappa}_1\Big], \qquad \boldsymbol{\kappa}_\beta = \sum_{j\geq 2} \frac{\boldsymbol{v}_j^\top \partial_{\boldsymbol{v}}\gamma}{\lambda_1 - \lambda_j}\boldsymbol{\kappa}_j, \quad \partial_{v_k}\gamma = \int (\boldsymbol{A}\boldsymbol{\varepsilon}_r)^\top \partial_{v_k}\boldsymbol{m}. \tag{33}$$

The *adjoint curvature field* $\boldsymbol{\kappa}_\beta$ collapses every eigenvector term into one field.

Mid-plate value. In the same way, $\delta w_c = \int \boldsymbol{\kappa}_{\mathrm{mid}}^\top \delta\boldsymbol{D}\,\boldsymbol{\kappa}_1$, with $\boldsymbol{\kappa}_{\mathrm{mid}} = \sum_{j\geq 2} \frac{w_c(\boldsymbol{v}_j)}{\lambda_1 - \lambda_j}\boldsymbol{\kappa}_j$.

Detuning. Since $J = \gamma/(\omega_1^2 w_c^2)$,

$$\delta J = \frac{\delta\gamma}{\omega_1^2 w_c^2} - \frac{\gamma\,\delta\omega_1^2}{\omega_1^4 w_c^2} - \frac{2\gamma\,\delta w_c}{\omega_1^2 w_c^3}, \qquad \frac{\partial J}{\partial\hat{\boldsymbol{\theta}}} = \boldsymbol{\Phi}^\top \Big[w_q \frac{\partial J}{\partial\theta_q}\Big]_q, \tag{34}$$

where w_q are the quadrature weights. The whole gradient costs one eigensolve of dimension N_w, one in-plane solve of dimension $2N_s$, and pointwise contractions. It does not depend on M, so the support grid can be refined freely. Equation (34) is the local optimality condition of Proposition 5, plus the mode-shape corrections carried by $\boldsymbol{\kappa}_\beta$ and $\boldsymbol{\kappa}_{\mathrm{mid}}$.

4.4 Verification of the evaluator and its gradient

- **Gradient.** Directional finite differences agree with (34), and with the ω_1 and γ derivatives, to 10^{-9}–10^{-7} relative.
- **Isotropy.** With $\boldsymbol{Q}$ isotropic, $\partial J/\partial\theta$ vanishes to 10^{-10} relative (Corollary 2).
- **Proposition 1.** $\int e_G = \gamma$ and $\int e_K = \omega_1^2$ hold to 10^{-10}.
- **Rotation.** The rotation (1) reproduces the classical $\bar{\boldsymbol{Q}}$ to machine precision, and Proposition 4 holds to 10^{-9} for random strains.
- **Uniform path.** The fast uniform-laminate path equals the general one to 10^{-13}.
- **Reference values.** For the UD plate the evaluator gives $f_1 = 256.3\,\mathrm{Hz}$ and a 10.95% shift at $a/h = 0.4$, in agreement with the 10.96% of the assumed-modes model of the finite-element comparison (Appendix C).

Basis convergence is in Table 1. All checks are in `tests/test_relaxed_plate.py`.

5 Results

5.1 The one-dimensional test (P1, P2)

Uniform invariance. Rotating every ply of the symmetric beam uniformly to 20° or 45° lowers the linear frequency to 0.895 and 0.561 of the UD value (207.2 Hz). On the full FE model, the cubic SSM detuning coefficient is unchanged to four digits (ratio 1.0000), as Corollary 3 requires. No uniform layup can therefore change J at all.

Table 1: Convergence of the evaluator on the reported held-frequency steered design (generated by `conv_table.py`). J/J_{UD} and the relaxed share are converged on the bold basis; three in-plane modes per direction under-resolve the steered relaxation. The steered frequency is a Ritz upper bound that falls by 0.9% as the transverse basis grows (the UD frequency does not change), so "held frequency" holds to about 0.9% across bases and 1.4% against the independent FE model. Bold: the basis used throughout.

transverse $m, n \leq$	in-plane $i, j \leq$	$f_{1,\mathrm{UD}}$ (Hz)	f_1 (Hz)	J/J_{UD}	relaxed share (UD)	relaxed share (steered)
8	3	256.3	256.3	0.523	33%	31%
8	4	256.3	256.3	0.420	33%	44%
8	5	256.3	256.3	0.375	33%	50%
8	**6**	**256.3**	**256.3**	**0.366**	**33%**	**51%**
8	8	256.3	256.3	0.360	33%	52%
12	6	256.3	254.6	0.370	33%	51%
12	8	256.3	254.6	0.364	33%	52%
16	8	256.3	253.9	0.366	33%	51%

Steering. Table 2 and Fig. 2 report the optimized symmetric beams. Every design has $\boldsymbol{B} \equiv \mathbf{0}$, so the bending–extension coupling plays no role. The optimizer does exactly what Proposition 6 predicts: it turns the fibers off-axis in two zones centered, to within the knot spacing of the angle field ($x/L = 0.25$, 0.75), on the inflection points ($x/L = 0.224, 0.776$), where φ''^2 vanishes, and keeps them axial where the beam bends (Fig. 2b). The SSM of the full FE beam confirms the condensed evaluator's prediction for every design to within 0.011 in J/J_{UD}.

- **Tow-curvature limits.** At the 900°/m limit and $\omega_1 \geq 0.98\,\omega_{1,\mathrm{UD}}$, the detuning falls by 19%. At 1800°/m it falls by 34%, and with no tow limit by 59%.
- **Frequency allowance.** Relaxing the frequency to $0.90\,\omega_{1,\mathrm{UD}}$ gives 38%, 56% and 62% respectively.
- **The coupling alternative.** The intuitive alternative is an unsymmetric laminate whose bending–extension coupling $\boldsymbol{B}(x)$ injects a softening quadratic force. We optimize it with its own, related objective (the RMS backbone drift over $a/h \leq 0.4$; two independently steered plies with two sine modes each, the same tow limit, and $\omega_1 \geq 0.90\,\omega_{1,\mathrm{UD}}$). It reduces that drift by only 2%, at $0.90\,\omega_{1,\mathrm{UD}}$. The comparison is therefore not like-for-like in objective, but the gap (a few percent against 38% for symmetric steering at the same 900°/m limit and frequency floor) is far larger than the difference between the two objectives.

The lever on a clamped beam is therefore the spatial separation of curvature and stretch, *not* the bending–extension coupling. Its size is set by how narrow the compliant zones can be (Corollary 4). With no tow limit the two zones are each $\approx 0.05L$ wide at near-transverse fibers (total compliant length $\ell \approx 0.1L$), and (26) gives $J/J_{\mathrm{UD}} \approx 0.4$–$0.5$, against the computed 0.405. At 900°/m a 90° turn needs a third of the span, so the zones cannot both be narrow and reach low modulus. This is why that limit binds.

5.2 The achievable frequency–hardening set of the plate (P3)

Figure 3a places every design class in the plane ($\omega_1/\omega_{1,\mathrm{UD}}$, J/J_{UD}) at the same mass.

Uniform layups. A uniform single angle traces a one-parameter curve. As θ grows from 0° the plate stiffens (up to $1.52\,\omega_{1,\mathrm{UD}}$ at 90°, fibers across the short span), and J first falls to 0.57 at

Table 2: Symmetric steered beam: detuning ratio J/J_{UD} from the condensed evaluator (optimizer) and from the full FE + cubic SSM (confirmation), with the realized frequency ratio.

tow limit	frequency floor	J/J_{UD} condensed	J/J_{UD} FE+SSM	$\omega_1/\omega_{1,\mathrm{UD}}$ (SSM)
900°/m	0.98	0.818	0.813	0.979
900°/m	0.90	0.627	0.621	0.898
1800°/m	0.98	0.667	0.656	0.978
1800°/m	0.90	0.430	0.440	0.896
none	0.98	0.414	0.405	0.976
none	0.90	0.370	0.375	0.916

$\theta = 37°$, where $\omega_1 = 1.28\,\omega_{1,\mathrm{UD}}$, then rises. The 20,736 uniform symmetric stacks on a 15° grid fill a region with a global minimum of 0.50 at $1.55\,\omega_{1,\mathrm{UD}}$. A grid envelope is not the strongest uniform competitor: between grid points the continuous single angle already beats it (0.75 vs. 0.81 at $1.1\,\omega_{1,\mathrm{UD}}$). We therefore also optimize the four ply angles of $[\theta_4/\theta_3/\theta_2/\theta_1]_s$ *continuously* at each held frequency (SLSQP with the frequency equality, 22 starts from the best grid stacks, the single-angle solution and random stacks; `stack_continuous.py`). The best uniform laminate reaches $J/J_{\mathrm{UD}} = 0.74$, 0.61 and 0.55 at 1.1, 1.2 and $1.3\,\omega_{1,\mathrm{UD}}$.

The UD floor. Corollary 1 proves, and the enumeration confirms, that the UD plate is the *lowest-frequency* member of the whole uniform family: the smallest stack frequency is $1.000\,\omega_{1,\mathrm{UD}}$. At exactly $\omega_1 = \omega_{1,\mathrm{UD}}$ the continuous stack optimum is UD itself (all ply angles $\to 0$); the best grid stack within ±1% of that frequency, at a slightly different ω_1, has $J/J_{\mathrm{UD}} = 0.997$, a negligible difference. For a plate mandated to the UD frequency, the uniform design space is therefore a single point.

Steered designs. At exactly $\omega_{1,\mathrm{UD}}$, the steered single ply reaches $J/J_{\mathrm{UD}} = 0.366$, a 63% reduction, with $R_{\min} = 20\,\mathrm{mm}$ imposed (it is active but changes J only in the fourth digit: without it the optimum is 0.366, Sec. 5.4). The steered designs lie well below the best uniform laminate at *every* frequency we test: 0.29, 0.24 and 0.21 at 1.1, 1.2 and $1.3\,\omega_{1,\mathrm{UD}}$, against 0.74, 0.61 and 0.55 for the best continuous stacks. Steering can also *lower* the frequency below the uniform floor (0.55 at $0.9\,\omega_{1,\mathrm{UD}}$), which no uniform laminate can do. Steering the best stack at $\omega_{1,\mathrm{UD}}$ gives 0.37, no better than the steered single ply. For this plate and objective, the in-plane channel of Sec. 3.3 dominates the through-thickness channel. Prediction P3 holds with a wide margin.

5.3 The mechanism (P4)

Figure 4 tests the mechanism of Sec. 3 directly.

- **Where the energies live.** Both energies of the UD mode peak at the center. The bending energy is concentrated there, with the principal curvature along y, across the short span (panel a). The relaxed-stretch energy lies in a band along x through the center, with its across-stretch direction along x there and turning toward the corners (panel b).
- **Sign of the trade-off.** At the held UD frequency the multiplier is negative, $\Lambda = -0.16$ (dimensionless: e_G and e_K normalized by the UD γ and ω_1^2): minimizing the stretch energy alone would raise the frequency to $1.20\,\omega_{1,\mathrm{UD}}$, so the constraint pushes the fibers *across* the curvature. At the center the two preferences agree (fibers along x), and the pointwise design $\theta^\star$ keeps the UD direction there.

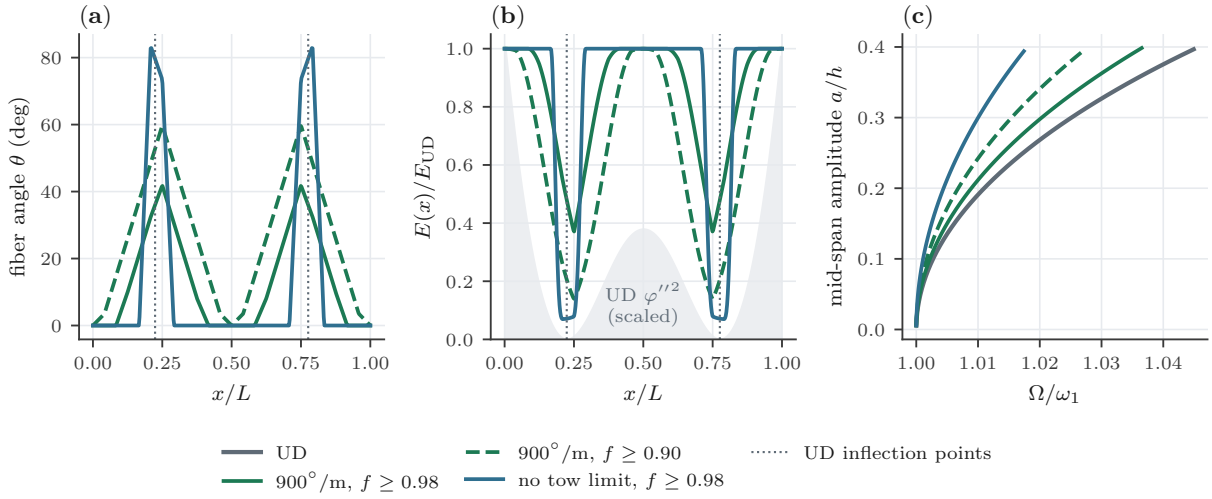


Figure 2: The one-dimensional test. **(a)** Optimized mirror-symmetric fiber angle of the symmetric ($\boldsymbol{B} = \boldsymbol{0}$) clamped beam for three designs: tow limit 900°/m with the frequency held to $f \geq 0.98$ or $f \geq 0.90$ of the UD value, and no tow limit with $f \geq 0.98$. The UD beam ($\theta \equiv 0$, $E = E_{\mathrm{UD}}$) appears in (c) only. **(b)** The axial modulus these fields produce, against the UD curvature weight φ''^2 that sets ω_1^2 (shaded). Fibers turn compliant where the beam does not bend: the inflection points (dotted). **(c)** Mid-span backbones from the full FE beam and its cubic SSM; steering flattens the backbone markedly. (The optimized unsymmetric, $\boldsymbol{B}$-coupled laminate was computed on a different beam model and objective, so it is compared in the text, not plotted.)

- **Conflict.** Away from the center the across-stretch and across-curvature directions separate (panel c); their energy-weighted mean angle is 33°. This is where a single angle must compromise.
- **Pointwise prediction.** The first-order design $\theta^\star(\boldsymbol{x})$ of Proposition 5, computed from the UD fields alone with no optimizer and no smoothness (panel d), already gives $J/J_{\mathrm{UD}} = 0.46$ when evaluated on the full model. Its frequency is $0.935\,\omega_{1,\mathrm{UD}}$, because the frozen-field model ignores the adaptation of the mode.
- **Optimized field.** The optimized, curvature-limited field (panel e) shares the large-scale pattern of $\theta^\star$ (fibers turned off-axis toward the short edges) but departs from it in detail: the energy-weighted RMS angle difference is 29° over the plate and 28° where the energies are large, and at the energy peak the optimum is tilted to −21°. The frozen-field design ignores the adaptation of the mode and the smoothness and tow limit of a manufacturable field, both of which the optimizer must respect.

Panel f shows which terms move, normalized by UD:

- the raw (restrained) membrane cubic falls to $k_{nl} = 0.52$;
- the relaxation $2\boldsymbol{g}^\top \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$ changes to 0.81 of its UD value, a share of k_{nl} rising from 33% to 51%;
- the relaxed cubic falls to $\gamma = 0.38$;
- the frequency is held ($\omega_1^2 = 1.00$) and the mode's center value changes to $w_c = 1.01$.

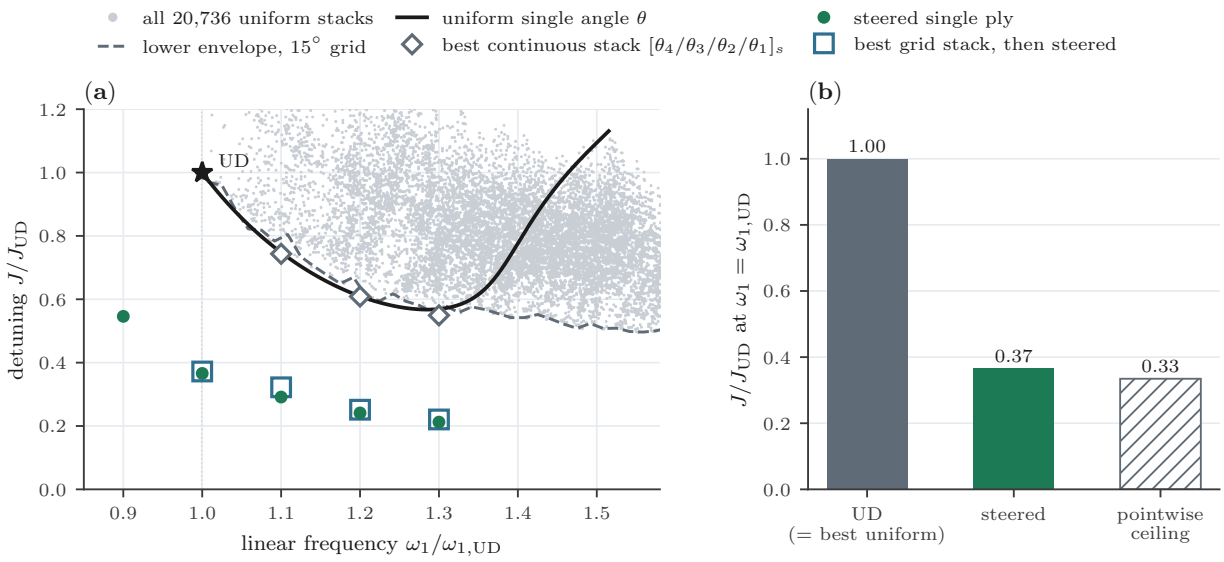


Figure 3: What each laminate freedom can reach, at fixed mass. **(a)** Linear frequency vs. frequency-normalized hardening. Black: uniform single angle. Gray dots: 20,736 uniform symmetric stacks on a 15° grid (cloud clipped at $J/J_{\mathrm{UD}} = 1.2$ and $\omega_1/\omega_{1,\mathrm{UD}} = 1.58$), with their lower envelope dashed; open diamonds: the best *continuous* stack at each held frequency, the strongest uniform competitor. Dotted vertical line: $\omega_1 = \omega_{1,\mathrm{UD}}$. Green: steered single ply, optimized at five target frequencies with $R_{\min} = 20\,\mathrm{mm}$. Blue squares: the best 15°-grid stack at each target, then steered (where the two coincide, the green dot sits inside the square; at $0.9\,\omega_{1,\mathrm{UD}}$ no uniform stack exists, so there is no square). The UD plate (star) is the lowest-frequency uniform laminate. **(b)** At $\omega_1 = \omega_{1,\mathrm{UD}}$: UD, which is also the best uniform laminate there (Corollary 1), the steered single ply, and the unmanufacturable pointwise ceiling of Sec. 5.4. The best stack at this frequency is UD to within 0.3% in J ($[0/0/0/15]_s$), and steering it gives 0.37, so it is not shown separately.

The hardening falls mainly through the direct term of (13): fibers across the principal stretch store less membrane energy for the same stretch, so k_{nl} falls to 0.52 of its UD value. The relaxation falls less (to 0.81), so its share of the stretch energy rises and γ falls further than k_{nl} alone. Prediction P4 holds partly: $\theta^\star$ alone captures most of the gain, though at $0.935\,\omega_{1,\mathrm{UD}}$, but the optimum departs from it by 28° RMS even where the energies are large.

5.4 What bounds the lever (P5)

Anisotropy. Along the family $\boldsymbol{Q}(s) = \boldsymbol{Q}_{\mathrm{iso}} + s(\boldsymbol{Q} - \boldsymbol{Q}_{\mathrm{iso}})$, which keeps the isotropic invariants U_1, U_4, U_5 and scales U_2, U_3 by s, the held-frequency steered optimum falls monotonically with the anisotropy:

- $J/J_{\mathrm{UD}} = 0.98$ at $E_1/E_2 = 1.2$;
- 0.84 at 2.9;
- 0.37 at 14.6 (CFRP);
- 0.21 at 31.9 (Fig. 5a).

The isotropic limit has no lever (Corollary 2).

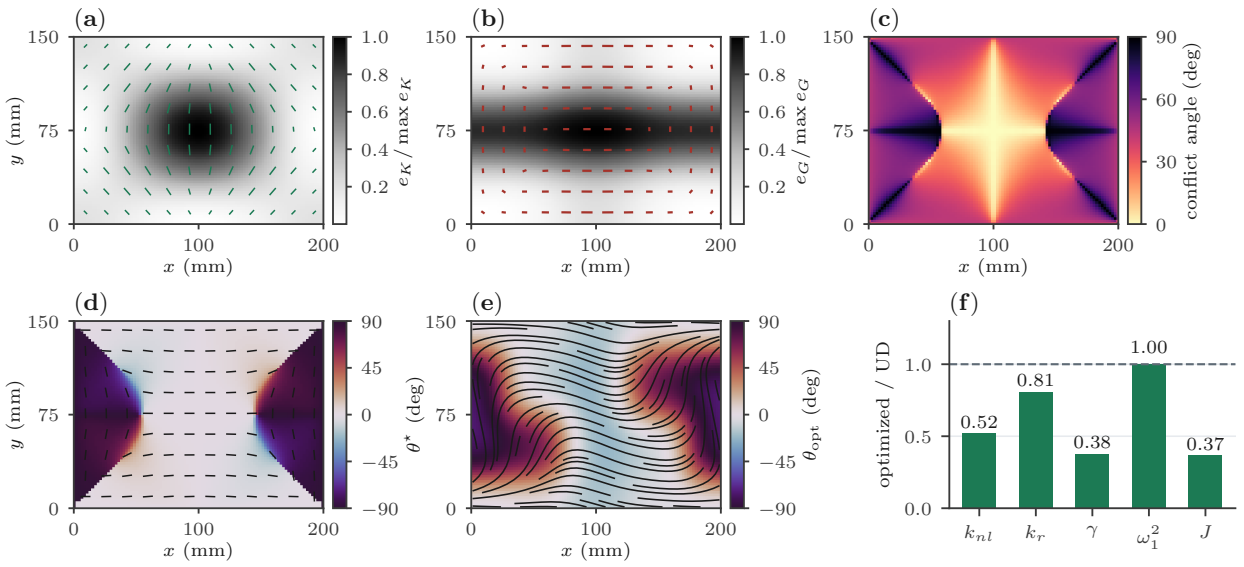


Figure 4: The mechanism. **(a)** Bending-energy density e_K of the UD fundamental (it integrates to ω_1^2), with the fiber direction that locally maximizes it (principal curvature; green ticks, length $\propto$ anisotropic amplitude). **(b)** Relaxed-stretch density e_G (integrates to γ), with the direction that locally minimizes it (across the principal stretch; red ticks). **(c)** Conflict angle between the two preferred directions at the held frequency ($\Lambda < 0$): across the stretch, $\varphi_G + 90°$, and across the curvature, $\varphi_K + 90°$, i.e. the angle between the red ticks of (b) and the green ticks of (a) rotated by 90° (the green ticks show the curvature direction φ_K itself, so at the center they are perpendicular to the red ticks and the conflict is zero). The sharp ridges in (c) are lines where one of the two principal directions is undefined (equal principal strains). **(d)** First-order pointwise design $\theta^\star$ from the UD fields alone (Proposition 5); black ticks: the same direction field; the cyclic color map is lightest at $\theta = 0°$, so the pale regions are where $\theta^\star$ keeps the fibers along x. **(e)** Optimized curvature-limited field and its tow paths. **(f)** Optimized/UD ratios of the terms of (13), the raw cubic k_{nl} and the relaxation $k_r = 2\boldsymbol{g}^\top \boldsymbol{K}_{\mathrm{in}}^{-1}\boldsymbol{g}$, of their difference γ, and of ω_1^2 and J; dashed: parity with UD. All maps show the full 200 mm × 150 mm plate; angles are measured from the x axis.

Tow curvature. The held-frequency optimum degrades smoothly as the minimum tow radius grows:

- 0.39 at $R_{\min} = 30$ mm;
- 0.47 at 50 mm;
- 0.57 at 80 mm;
- 0.71 at 150 mm;
- 0.84 at 400 mm (Fig. 5b).

Below $R_{\min} \approx 20$ mm it saturates at 0.37. Two checks show that the saturation is physical rather than an artifact of the 50 mm support grid.

- **Finer grid.** Refining the grid to 33 mm and 25 mm changes the optimum only to 0.361 and 0.358 at $R_{\min} = 20$ mm, and to 0.358 and 0.354 with no tow limit.

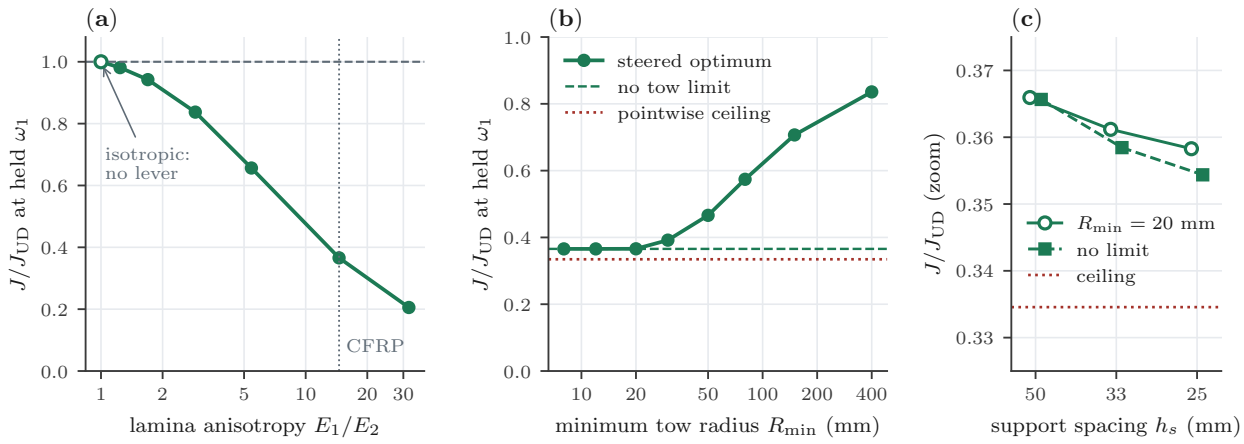


Figure 5: What bounds the lever, at held frequency. **(a)** Steered optimum vs. lamina anisotropy along a family that scales only the anisotropic invariants ($E_1/E_2 = 1$ is isotropic, where $J/J_{\mathrm{UD}} = 1$ exactly by Corollary 2, so that point is not computed; dashed: parity with UD; dotted: CFRP). **(b)** Manufacturability front: optimum vs. minimum tow radius on the 50 mm support grid; dashed: the same grid with no tow limit; dotted: the pointwise (unmanufacturable) ceiling. **(c)** Support-grid refinement (zoomed ordinate; the three support spacings are plotted as equally spaced categories, with the two series offset slightly for visibility): at $R_{\min} = 20$ mm (open) and with no tow limit (filled) the optimum moves by less than 0.02 and approaches the ceiling, so the saturation in (b) is physical, not a parametrization artifact.

- **Pointwise ceiling.** Freeing θ at every quadrature point, with no smoothness and no tow limit (an unmanufacturable bound on single-ply steering), gives 0.33.

The plate's strain fields vary on a length scale of a few centimeters. Once tows can turn on that scale, further manufacturing freedom buys little. Prediction P5 holds: the lever is set by the material anisotropy and by the ratio of the tow radius to the length scale of the mode's strain fields.

5.5 Independent confirmation on the full model

The design was produced on the condensed, adapted-mode evaluator. We confirm it on the full multi-mode plate: 36 transverse and 72 in-plane modes, in-plane inertia retained, an independent Gauss-quadrature assembly, and amplitudes measured as the true mid-plate deflection (Table 3).

- **Full SSM.** The full cubic SSM gives detuning coefficients β within 1.2% of the evaluator for both designs. The detuning ratio is 0.359, a 64% reduction.
- **EPMC.** Full-order EPMC continuation ($H = 3$ harmonics) gives a ratio of 0.362.
- **Frequency.** The steered design's full-model frequency is 258.3 Hz against 256.3 Hz for UD. The difference is the Ritz upper-bound convergence of the transverse basis for the steered field (Table 1: the steered frequency falls by 0.9% from $m, n \leq 8$ to $m, n \leq 16$ while the UD frequency is unchanged), not a design change.
- **Discretization.** An independent finite-element model (ACM Kirchhoff bending with Q4 membrane elements, Appendix C) evaluated on the optimized field gives a detuning ratio of

Table 3: Confirmation of the held-frequency design. β: small-amplitude coefficient of $\Omega/\omega_1 - 1 = \beta(a/h)^2$ at the plate center.

model	f_1 UD (Hz)	f_1 steered (Hz)	β UD	β steered
design evaluator (64+72 modes, condensed)	256.3	256.3	0.684	0.250
full plate, cubic SSM (36+72 modes)	256.3	258.3	0.689	0.248
full plate, EPMC ($H = 3$)	—	—	0.680	0.246
independent FE (ACM + Q4, 48×36, App. C)	256.2	252.6	0.685	0.250

0.364. Across the three discretizations the steered design's frequency lies within $\pm 1.4\%$ of the UD value, which is the accuracy to which "held frequency" is meaningful here.

5.6 Consequence: a wider jump-free range

Proposition 3 predicts that the held-frequency reduction widens the jump-free amplitude by $(J_{\text{steered}}/J_{\text{UD}})^{-1/2} = 1.67$. We compute a_c numerically as the drive at which the full-order damped SSM forced response ($\zeta = 2\%$, center point load) first becomes multivalued. It rises from $a_c/h = 0.208$ to 0.347, a factor 1.66 in amplitude and 1.64 in drive. This agrees with the predicted ratio, and the absolute thresholds agree too: (16) with the full-model β gives 0.211 and 0.353.

Direct time integration then shows the jump itself (Fig. 6b,c). We drive both panels with the same point load, set between their two thresholds, and sweep the frequency up and down. The UD panel is hysteretic: the up and down sweeps differ by up to $0.128\,h$ across the fold. The steered panel, at the same frequency and mass, is single-valued: its sweeps coincide to $0.000\,h$ and follow the SSM forced response. A finer sweep (41 steps over $\Omega/\omega_1 \in [1.00, 1.08]$; `confirm_fine_sweep.py`) confirms both: the UD up and down sweeps differ at 2 consecutive steps across the fold (by up to $0.125\,h$), the steered sweeps at none.

6 Discussion

6.1 Answer to the question

- **Can steering do it?** Yes. At the frequency (held exactly in the design model, to within 1.4% on refined bases and an independent finite-element model) and mass of the unidirectional plate, a manufacturable steered tow reduces the frequency-normalized hardening by 63%. No uniform layup can reach that frequency at a lower hardening, and steering undercuts the best stacking sequences at every higher frequency we test.
- **Why?** The frequency and the hardening read the same rotated stiffness through two strain fields whose principal directions turn differently over the plate, and an optimal design must satisfy the trade-off between them pointwise (Proposition 5).
- **What bounds it?** The material anisotropy (U_2, U_3), and the tow radius relative to the length scale of the fields.
- **How does it compare?** Steering is the stronger channel here. It is also the one that does not require changing the bill of materials.

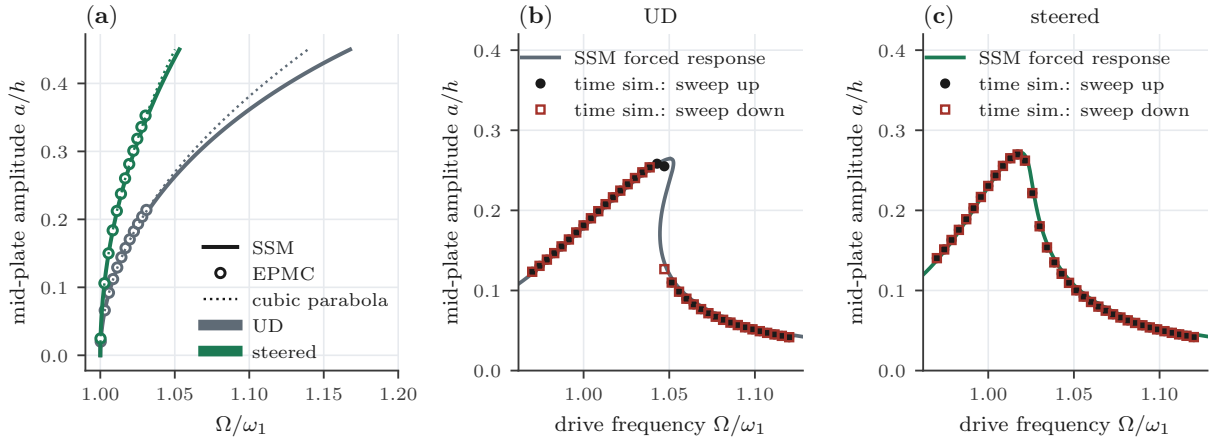


Figure 6: Confirmation and consequence. **(a)** Mid-plate backbones of the full multi-mode plate: cubic SSM (lines) and full-order EPMC (markers), UD (gray) and the held-frequency steered design (green). Dotted: the cubic parabola $1 + \beta(a/h)^2$ with the full-model SSM β (within 1.2% of the condensed evaluator's). **(b,c)** Time-integrated harmonic sweeps (up: black dots; down: red squares; 300 forcing periods per step, and a point is kept only once its amplitude is steady to 0.4%, which excludes one UD point at the fold) at one drive between the two jump thresholds, over the damped SSM forced response. The UD panel folds and jumps. The steered panel, at the same linear frequency and mass, stays single-valued.

6.2 What is new

- **Design variable.** Reduced-order optimization of nonlinear dynamics [16, 15, 17, 18, 19] has used shape, size, density or other system parameters. We use a manufacturable fiber-orientation field.
- **Objective.** Variable-stiffness composites [23, 26] have been steered for linear objectives and for postbuckling [33, 34], and for the linear frequency with the nonlinear response examined afterwards [31, 32]. We steer them for the hardening itself, at fixed linear frequency.
- **Mechanism, not only an optimum.** The paper's main contribution is the mechanism: a local Fourier theory that predicts when steering must help, where the fibers should go, and what limits the gain, tested prediction by prediction.
- **Analysis versus design.** Nonlinear vibrations of variable-stiffness plates have been computed for prescribed fiber paths [39, 40, 27, 28, 29, 30]. Here the path is designed at a held frequency, and the reason it works is identified.
- **One-dimensional result.** On a clamped beam the operative lever is not the bending–extension coupling but the harmonic-versus-curvature-weighted mean of Proposition 6, which requires no coupling at all.

6.3 Limitations and scope

- **Geometry and support.** Results are for a simply supported, immovable-edge rectangular plate. The mechanism is general, but the numbers depend on where the curvature and the stretch live, which other supports change.

Table 4: Status of the predictions.

prediction	evidence	verdict
P1 isotropic material has no lever	Cor. 2; gradient $\equiv 0$ to 10^{-10}; Fig. 5a	holds (exact)
P2 uniform angle cannot change a single-ply beam; steering can, at the inflection points	SSM ratio 1.0000; Tab. 2; Fig. 2	holds
P3 steered plate reaches (ω_1, J) no uniform layup reaches	Fig. 3; 20,736 stacks	holds
P4 optimized field follows preferred directions; $\theta^\star$ captures most of the gain	Fig. 4; $\theta^\star$: 0.46 at $0.935\,\omega_{1,\mathrm{UD}}$; optimum departs by 28° RMS where energies are large	holds partly
P5 lever grows with anisotropy, saturates at the field length scale	Fig. 5	holds
consequence: a_c widens as $J^{-1/2}$	1.66 vs. 1.67; time-domain sweeps	holds

- **Constant thickness.** We assume constant thickness and mass. Real AFP steering produces tow gaps or overlaps whose thickness and stiffness changes are not modeled. Including them, through a thickness field tied to $\theta(\boldsymbol{x})$, is the main step toward a buildable design.
- **Amplitude range.** The design quantity is the small-amplitude cubic coefficient. The cubic SSM reproduces the full-order EPMC detuning to within 5% over the amplitudes EPMC reached ($a/h \leq 0.21$ for UD, ≤ 0.35 for the steered design), and the jump-onset amplitudes reported lie inside these ranges. Higher amplitudes need the quintic SSM or EPMC (Appendix D).
- **Single mode, no internal resonance.** On the full multi-mode plate the steered design's lowest frequencies are in ratios 1:2.49:2.86:4.16 (`mode_ratios.py`). The nearest commensurability (3:1 with the third mode) is detuned by 5%, and that mode is antisymmetric about the center. The six-mode modal time-integration model contains these modes and would show an internal-resonance exchange; its sweeps follow the single-mode forced response instead (Fig. 6b,c).
- **Optimality.** The optimizer is local (multi-start SLSQP). The pointwise ceiling of Sec. 5.4 bounds what any single-ply field can do. Stacks have eight plies (four independent angles), optimized continuously at each held frequency; more plies or a lamination-parameter search could lower the uniform optimum further; the steered values are at most 40% of the present one. The UD floor at $\omega_{1,\mathrm{UD}}$ holds for every uniform laminate (Corollary 1).
- **Damping.** Damping is fixed (2% of critical, mass-proportional) and not designed.
- **Idealized laminate.** The plate designs are a single 2 mm steered ply, i.e. an unbalanced laminate, and tow gaps and overlaps are ignored. The beam design with no tow limit turns to about 83° in narrow zones of the 20 mm-wide strip.

6.4 Outlook

The same reduction and adjoint carry over directly to three extensions:

- *prescribing* the backbone (a target frequency–amplitude law) instead of minimizing the hardening;
- adding a density field, so that topology and tow path are designed together;

- steering stability boundaries, such as supersonic panel flutter, whose nonsymmetric aeroelastic operator enters the same first-order pencil; the flutter and post-flutter response of plates with prescribed curvilinear fibers has been analyzed [35, 36, 37], but not designed.

The mechanism of Sec. 3 predicts where each extension should help: wherever two functionals of the same stiffness are weighted by different strain fields.

7 Conclusions

1. The frequency and the geometric hardening of a laminated panel integrate the same rotated stiffness against the curvature and the relaxed stretch of the fundamental. Each density depends on the local fiber angle only through U_2, U_3 and the local principal directions (Proposition 4).
2. A uniform single angle cannot change the frequency-normalized hardening of a single-ply beam. Symmetric steering cuts it by up to 59% with the frequency within 2.4% of the UD value, by placing compliant fibers at the inflection points. In our comparison bending–extension coupling was a minor lever (2% on its own drift objective, against 38% for symmetric steering at the same 900°/m tow limit and frequency floor); a richer coupled parametrization might do better.
3. On a simply supported plate, a curvature-feasible steered field reduces the frequency-normalized hardening by 63% at the unidirectional frequency (held exactly in the design model, to within 1.4% on refined bases and an independent finite-element model) and mass. This point is unreachable by any uniform layup, and the steered designs undercut the best stacking sequences at every frequency tested.
4. A pointwise design from the UD fields alone captures most of the gain; the optimized field shares its large-scale pattern but not its detail. Steering lowers the restrained stretch energy k_{nl} to 0.52 of its UD value; the relaxation falls less (0.81), so its share of the stretch energy rises (33% → 51%) and γ falls further than k_{nl}.
5. The lever vanishes for isotropic material, grows monotonically with anisotropy, and saturates once the minimum tow radius is below about 20 mm.
6. The full-order model confirms the reduction (SSM, EPMC, time integration), and the jump-free amplitude widens by 1.66×, in agreement with $a_c \propto J^{-1/2}$.

Data and code availability. The code, result files and the scripts that generate every number, table and figure (Appendix E) are in the repository `https://github.com/jtempelman95/NLTopt` (branch `paper-revision`; see `docs/paper/README_REPRODUCE.md`). The SSM, EPMC and time-integration scripts build on the author's harmonic-balance package PyHB, which is not public and is available on request; the design evaluator and the regeneration of every number from the committed result files do not need it.

Competing interests. The author declares no competing interests.

A Computation of the SSM coefficients

This appendix gives the exact recursion that produces every coefficient $\boldsymbol{w_m}$ (the geometry) and γ_{2k+1} (the reduced dynamics) used in the paper, matching the implementation line for line.

A.1 Master spectral data

Cast (9) in first order, $\boldsymbol{\mathcal{B}}\dot{\boldsymbol{z}} = \boldsymbol{\mathcal{A}}\boldsymbol{z} + \boldsymbol{F}(\boldsymbol{z})$, with $\boldsymbol{\mathcal{A}}, \boldsymbol{\mathcal{B}}$ as in Sec. 2.3. Solve the linear generalized eigenproblem $\boldsymbol{K}\boldsymbol{\varphi} = \omega_0^2 \boldsymbol{M}\boldsymbol{\varphi}$, mass-normalize $\boldsymbol{\varphi}^\top \boldsymbol{M}\boldsymbol{\varphi} = 1$, and set $\zeta_0 = \boldsymbol{\varphi}^\top \boldsymbol{C}\boldsymbol{\varphi}/(2\omega_0)$. The master pencil eigenpair solves $\boldsymbol{\mathcal{A}}\boldsymbol{v}_1 = \lambda_1 \boldsymbol{\mathcal{B}}\boldsymbol{v}_1$ with $\lambda_1 = -\zeta_0\omega_0 + i\omega_0\sqrt{1-\zeta_0^2}$; scale $\boldsymbol{v}_1$ so its displacement block is $\boldsymbol{\varphi}$. For symmetric $\boldsymbol{M}, \boldsymbol{C}, \boldsymbol{K}$ the pencil is complex-symmetric, so the left eigenvector is $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$ (normalized $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}}\boldsymbol{v}_1 = 1$); $(\lambda_1, \boldsymbol{v}_1)$ is refined from the seed $([\boldsymbol{\varphi}; \lambda_1\boldsymbol{\varphi}], \lambda_1)$ by Rayleigh-quotient inverse iteration, $\boldsymbol{y} = (\boldsymbol{\mathcal{A}} - \lambda_1\boldsymbol{\mathcal{B}})^{-1}\boldsymbol{\mathcal{B}}\boldsymbol{v}_1$, $\lambda_1 \leftarrow \boldsymbol{v}_1^\top \boldsymbol{\mathcal{A}}\boldsymbol{v}_1 / \boldsymbol{v}_1^\top \boldsymbol{\mathcal{B}}\boldsymbol{v}_1$, which converges cubically. (A dense QZ factorization of the first-order pencil is used only for non-symmetric systems; for the laminated beam its master eigenvector lost up to ten digits at some ply angles.) Set $\boldsymbol{v}_2 = \bar{\boldsymbol{v}}_1$, $\lambda_2 = \bar{\lambda}_1$. Seed the coefficient tables at first order: $\boldsymbol{w}_{(1,0)} = \boldsymbol{v}_1$, $\boldsymbol{w}_{(0,1)} = \boldsymbol{v}_2$, $R^1_{(1,0)} = \lambda_1$, $R^2_{(0,1)} = \lambda_2$.

A.2 Right-hand side at each order

For a multi-index $\boldsymbol{m} = (m_1, m_2)$, $|\boldsymbol{m}| \geq 2$, the homological equation (10) has right-hand side $\boldsymbol{G}_{\boldsymbol{m}} = \boldsymbol{G}^{\mathrm{nl}}_{\boldsymbol{m}} - \boldsymbol{G}^{\mathrm{cr}}_{\boldsymbol{m}}$, both assembled from *lower* orders. The nonlinear part is the order-$\boldsymbol{m}$ term of $-\boldsymbol{F}(\boldsymbol{W}(\boldsymbol{p}))$, a multi-index convolution of the displacement blocks $\boldsymbol{w}^x$ over all non-trivial partitions,

$$\boldsymbol{G}^{\mathrm{nl}}_{\boldsymbol{m}} = -\sum_{\substack{\boldsymbol{a}+\boldsymbol{b}=\boldsymbol{m} \\ \boldsymbol{a},\boldsymbol{b}\neq\boldsymbol{0}}} \begin{bmatrix} \boldsymbol{B}_2(\boldsymbol{w}^x_{\boldsymbol{a}}, \boldsymbol{w}^x_{\boldsymbol{b}}) \\ \boldsymbol{0} \end{bmatrix} - \sum_{\substack{\boldsymbol{a}+\boldsymbol{b}+\boldsymbol{c}=\boldsymbol{m} \\ \boldsymbol{a},\boldsymbol{b},\boldsymbol{c}\neq\boldsymbol{0}}} \begin{bmatrix} \boldsymbol{T}_3(\boldsymbol{w}^x_{\boldsymbol{a}}, \boldsymbol{w}^x_{\boldsymbol{b}}, \boldsymbol{w}^x_{\boldsymbol{c}}) \\ \boldsymbol{0} \end{bmatrix}, \tag{35}$$

and the cross-term part is what $\mathrm{D}\boldsymbol{W}\,\boldsymbol{R}$ already contributes from known nonlinear reduced coefficients, $\boldsymbol{G}^{\mathrm{cr}}_{\boldsymbol{m}} = \boldsymbol{\mathcal{B}} \sum_{j=1,2} \sum_{\boldsymbol{a}+\boldsymbol{e}_j=\boldsymbol{m},\, |\boldsymbol{b}|\geq 2} a_j\, \boldsymbol{w}_{\boldsymbol{a}} R^j_{\boldsymbol{b}}$, evaluated over the already-computed $\boldsymbol{w}, R$.

A.3 Solving each order

Sweep $q = 2, \ldots,$ order, and within each q the indices $\boldsymbol{m} = (m_1, m_2)$ with $m_1 \geq m_2$ (the conjugate partner is filled by symmetry). Put $\Lambda_{\boldsymbol{m}} = m_1\lambda_1 + m_2\lambda_2$. Two cases:

- **Near-resonant** ($m_1 - m_2 = 1$, i.e. $\boldsymbol{m} = (k{+}1, k)$, where $\Lambda_{\boldsymbol{m}} = \lambda_1$ conservatively): the operator $\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}}$ is (near-)singular along $\boldsymbol{v}_1$. Project onto the left eigenvector to read the reduced coefficient, then solve the deflated system for the geometry under the normal-form gauge $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}}\boldsymbol{w}_{\boldsymbol{m}} = 0$:

$$\boxed{\gamma_{2k+1} = R^1_{\boldsymbol{m}} = \boldsymbol{u}_1^* \boldsymbol{G}_{\boldsymbol{m}}}, \qquad (\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})\, \boldsymbol{w}_{\boldsymbol{m}} = \boldsymbol{G}_{\boldsymbol{m}} - \boldsymbol{\mathcal{B}}\boldsymbol{v}_1\, R^1_{\boldsymbol{m}}, \tag{36}$$

  followed by $\boldsymbol{w}_{\boldsymbol{m}} \leftarrow \boldsymbol{w}_{\boldsymbol{m}} - (\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}}\boldsymbol{w}_{\boldsymbol{m}})\boldsymbol{v}_1$. Under light damping $\Lambda_{\boldsymbol{m}} \neq \lambda_1$ exactly and the solve is nonsingular; at exact resonance one uses the least-squares pseudo-inverse and the same deflation.

- **Non-resonant**: set $R_{\boldsymbol{m}} = \boldsymbol{0}$ and invert directly, $\boldsymbol{w}_{\boldsymbol{m}} = (\Lambda_{\boldsymbol{m}}\boldsymbol{\mathcal{B}} - \boldsymbol{\mathcal{A}})^{-1}\boldsymbol{G}_{\boldsymbol{m}}$.

The reduced coefficients are $\{\gamma_{2k+1}\} = \{R^1_{(k+1,k)}\}$ (γ_3 at $k{=}1$, γ_5 at $k{=}2$, …), and the backbone is (11).

A.4 Worked coefficient: the cubic term

At order 3 the only near-resonant index is $\boldsymbol{m} = (2, 1)$. By (35) its forcing is the cubic force on the master eigenvectors plus the quadratic acting on the order-2 map, $\boldsymbol{G}_{(2,1)} = -\big[\, 3\,\boldsymbol{T}_3(\boldsymbol{v}_1, \boldsymbol{v}_1, \bar{\boldsymbol{v}}_1) +$

$2\boldsymbol{B}_2(\boldsymbol{v}_1, \boldsymbol{w}_{(1,1)}) + 2\boldsymbol{B}_2(\bar{\boldsymbol{v}}_1, \boldsymbol{w}_{(2,0)})\big]$ (displacement block), and (36) gives $\gamma_3 = \boldsymbol{u}_1^* \boldsymbol{G}_{(2,1)}$. For a single-DOF Duffing $\ddot{x} + \omega_0^2 x + \varepsilon x^3 = 0$ the quadratic vanishes, the order-2 map is zero, and γ_3 reduces to the classical $\Omega = \omega_0\big(1 + \frac{3}{8}(\varepsilon/\omega_0^2)A^2\big)$. For the von Kármán plate the membrane coupling $\boldsymbol{B}_2$ (the $u \leftrightarrow w^2$ term) is nonzero, its order-2 slaving $\boldsymbol{w}_{(1,1)}, \boldsymbol{w}_{(2,0)}$ enters through $\boldsymbol{B}_2$ above, and $\operatorname{Im}\gamma_3$ equals $\frac{3}{2\omega_1}$ times the relaxed cubic γ of Eq. (13), the direct membrane cubic *minus* the in-plane relaxation, up to the $O(4\omega_1^2/\omega_{\text{in}}^2)$ correction of Appendix B. The reduced coefficient (36), $\gamma_{2k+1} = \boldsymbol{u}_1^* \boldsymbol{G}_{(k+1,k)}$, is an adjoint projection, which is why its design sensitivity is inexpensive.

B The relaxed cubic is the SSM cubic (proof of Proposition 2)

Take (9) undamped with $\boldsymbol{B} = \boldsymbol{0}$ and split $\boldsymbol{x} = [\boldsymbol{q}; \boldsymbol{s}]$. The linear operators are block diagonal, so the master right eigenvector is $\boldsymbol{v}_1 = [\boldsymbol{\varphi}; \boldsymbol{0}; i\omega_1\boldsymbol{\varphi}; \boldsymbol{0}]$. Normalizing $\boldsymbol{u}_1^* \boldsymbol{\mathcal{B}} \boldsymbol{v}_1 = 1$ with $\boldsymbol{u}_1 = \bar{\boldsymbol{v}}_1$ gives the scaling $1/(2i\omega_1)$. The quadratic force has two parts. Its $\boldsymbol{s}$-rows are a symmetric form in $\boldsymbol{q}$, written $\boldsymbol{b}(\boldsymbol{q}, \boldsymbol{q})$ (the in-plane drive of the stretch). Its $\boldsymbol{q}$-rows are bilinear in $(\boldsymbol{q}, \boldsymbol{s})$. These are the two gradients of the single term $\int \boldsymbol{\varepsilon}_L^\top \boldsymbol{A} \boldsymbol{\varepsilon}_{NL}$. The cubic force acts in the $\boldsymbol{q}$-rows only.

Order 2. The forcing at $\boldsymbol{m} \in \{(2,0), (1,1), (0,2)\}$ has only $\boldsymbol{s}$-rows. Their coefficients are $1, 2, 1$ times $\boldsymbol{b}(\boldsymbol{\varphi}, \boldsymbol{\varphi})$, from expanding $(p_1 + p_2)^2$. The displacement-level homological equations are

$$(\boldsymbol{K}_{\text{in}} + \Lambda_{\boldsymbol{m}}^2 \boldsymbol{M}_{\text{in}})\, \boldsymbol{s}_{\boldsymbol{m}} = -c_{\boldsymbol{m}}\, \boldsymbol{b}(\boldsymbol{\varphi}, \boldsymbol{\varphi}), \qquad \Lambda_{(1,1)} = 0, \quad \Lambda_{(2,0)} = 2i\omega_1,$$

so $\boldsymbol{s}_{(1,1)} = -2\boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{b}$ is *exactly* static, while $\boldsymbol{s}_{(2,0)} = -(\boldsymbol{K}_{\text{in}} - 4\omega_1^2 \boldsymbol{M}_{\text{in}})^{-1}\boldsymbol{b}$.

Order 3. The resonant coefficient collects the cubic term ($3\times$) and the quadratic $\boldsymbol{q}$-rows acting on $\boldsymbol{s}_{(1,1)}$ and $\boldsymbol{s}_{(2,0)}$. With static $\boldsymbol{s}$, the combination is exactly $3(k_{nl} - 2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g})$, in which $\boldsymbol{s}_{(1,1)}$ supplies two thirds of the relaxation and $\boldsymbol{s}_{(2,0)}$ one third. Projecting with $\boldsymbol{u}_1$ gives $\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}\gamma$.

Error. Expanding the resolvent, $(\boldsymbol{K}_{\text{in}} - 4\omega_1^2 \boldsymbol{M}_{\text{in}})^{-1} = \boldsymbol{K}_{\text{in}}^{-1} + 4\omega_1^2 \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{M}_{\text{in}}\boldsymbol{K}_{\text{in}}^{-1} + \dots$. This perturbs the $(2,0)$ third of the relaxation by at most the relative amount $\rho/(1-\rho)$, with $\rho = 4\omega_1^2/\omega_{\text{in}}^2$. Hence

$$\operatorname{Im}\gamma_3 = \frac{3}{2\omega_1}(\gamma + \epsilon), \qquad |\epsilon| \le \tfrac{1}{3}\left(2\boldsymbol{g}^\top \boldsymbol{K}_{\text{in}}^{-1}\boldsymbol{g}\right)\frac{\rho}{1-\rho}.$$

For the plate, $\rho = 3.2 \times 10^{-3}$, so $|\epsilon|/\gamma < 0.1\%$. In-plane inertia is negligible for the backbone. The residual differences between (13) and the full SSM in Sec. 5.5 come from the transverse basis, not from the condensation.

C Finite-element validation of the assumed-modes discretization

Why this is needed. The reductions above are verified against the analytic Duffing backbone and against full-order EPMC and direct time integration. Those confirmations, however, are of the *same* assumed-modes system solved at higher fidelity: they validate the SSM *reduction*, not the assumed-modes *discretization*. A steered laminate is spatially heterogeneous and, off the principal axes, develops bend–twist couplings D_{16}, D_{26}; whether a handful of global sine modes resolve its response is a separate question. We settle it with an independent finite-element (FE) model and the identical backbone coefficient.

C.1 FE model

We take a flat, symmetric ($\boldsymbol{B} = \boldsymbol{0}$) plate on a uniform $n_x \times n_y$ rectangular mesh, bending and membrane discretized independently (they decouple linearly for $\boldsymbol{B} = \boldsymbol{0}$).

Bending (Kirchhoff). The non-conforming 12-DOF ACM rectangular element carries nodal degrees of freedom $\boldsymbol{d}_e = \{w,\ w_{,y},\ -w_{,x}\}$ at the four corners. On the element (local $s, t \in [0,1]$) the deflection is the 12-term polynomial

$$w(s,t) = \boldsymbol{p}^\top \boldsymbol{a}, \quad \boldsymbol{p} = [1, s, t, s^2, st, t^2, s^3, s^2t, st^2, t^3, s^3t, st^3]^\top; \tag{37}$$

enforcing the nodal DOF gives $\boldsymbol{d}_e = \boldsymbol{C}\boldsymbol{a} \Rightarrow \boldsymbol{a} = \boldsymbol{C}^{-1}\boldsymbol{d}_e$. With the curvature $\boldsymbol{\kappa} = [w_{,xx}, w_{,yy}, 2w_{,xy}]^\top = \boldsymbol{B}_b\boldsymbol{C}^{-1}\boldsymbol{d}_e$ ($\boldsymbol{B}_b$ the side-scaled second derivatives of $\boldsymbol{p}$), and 4×4 Gauss quadrature,

$$\boldsymbol{K}_e^b = \int_{\Omega_e} \boldsymbol{C}^{-\top}\boldsymbol{B}_b^\top\, \boldsymbol{D}(\theta_e)\, \boldsymbol{B}_b\boldsymbol{C}^{-1}\,\mathrm{d}\Omega, \quad \boldsymbol{M}_e^b = \rho h \int_{\Omega_e} \boldsymbol{C}^{-\top}\boldsymbol{p}\boldsymbol{p}^\top\boldsymbol{C}^{-1}\,\mathrm{d}\Omega, \tag{38}$$

$\boldsymbol{D}(\theta_e) = \frac{h^3}{12}\bar{\boldsymbol{Q}}(\theta_e)$ evaluated per element, so D_{16}, D_{26} are retained exactly.

Membrane (Q4). The in-plane fields use the 8-DOF bilinear quadrilateral; with the membrane strain–displacement $\boldsymbol{B}_m$ and $\boldsymbol{A}(\theta_e) = h\,\bar{\boldsymbol{Q}}(\theta_e)$, $\boldsymbol{K}_e^{\mathrm{in}} = \int_{\Omega_e} \boldsymbol{B}_m^\top \boldsymbol{A}(\theta_e)\boldsymbol{B}_m\,\mathrm{d}\Omega$.

Boundary conditions. Navier simple support: $w = 0$ on the boundary (bending) and $u = v = 0$ on the boundary (membrane) — matching the sine in-plane modes, which vanish on the edges. *Self-validation:* for a specially-orthotropic UD [0] plate the FE reproduces the closed form $\omega_{mn} = \pi^2\sqrt{[D_{11}(m/a)^4 + 2(D_{12}{+}2D_{66})(m/a)^2(n/b)^2 + D_{22}(n/b)^4]/\rho h}$ to $f_1 = 256.0$ vs. 256.3 Hz, converging monotonically with mesh (`examples/plate_fe_check.py`).

C.2 Backbone by static condensation on the mesh

We compute the *same* backbone coefficient the plate results use — Eq. (13) — but FE-discretized. Solve $\boldsymbol{K}^b\boldsymbol{\varphi} = \omega_1^2\boldsymbol{M}^b\boldsymbol{\varphi}$ for the fundamental mode (unit peak deflection); its von Kármán membrane strain is $\boldsymbol{\varepsilon}_{NL} = [\frac{1}{2}w_{,x}^2, \frac{1}{2}w_{,y}^2, w_{,x}w_{,y}]^\top$, with the gradients read off the ACM element. Assembling $\boldsymbol{g} = \int \boldsymbol{B}_m^\top\boldsymbol{A}\,\boldsymbol{\varepsilon}_{NL}$, $\mathrm{knl} = 2\int \boldsymbol{\varepsilon}_{NL}^\top\boldsymbol{A}\,\boldsymbol{\varepsilon}_{NL}$ and $K_b = \boldsymbol{\varphi}^\top\boldsymbol{K}^b\boldsymbol{\varphi}$, static condensation of the fast in-plane fields gives

$$\gamma = \mathrm{knl} - 2\,\boldsymbol{g}^\top(\boldsymbol{K}^{\mathrm{in}})^{-1}\boldsymbol{g}, \qquad \Omega(a)/\omega_1 = 1 + \beta\,(a/h)^2, \quad \beta = \tfrac{3}{8}\,(\gamma/K_b)\,h^2. \tag{39}$$

This costs *one* sparse solve — no continuation or time integration — because the condensation *is* the order-2 SSM slaving (App. A) evaluated on the mesh. The identical formula on the sine basis returns the assumed-modes backbone, so any difference is purely discretization (`examples/plate_fe_backbone.py`).

C.3 Results

Table 5 compares the two discretizations for a homogeneous and a strongly steered field ($\theta = 20^\circ + 45^\circ \sin\frac{\pi x}{a}\sin\frac{\pi y}{b}$, sweeping 20°–65°).

Two conclusions. *(i)* For a homogeneous laminate the FE and assumed-modes backbones agree to 1%, validating the FE nonlinear model (membrane, coupling, condensation). *(ii)* For strong steering the assumed-modes frequency is a Ritz upper bound that converges slowly, but the *normalized backbone detuning* $\beta \propto \gamma/K_b$ is far more robust — a ratio of two integrals of the *same* slightly-inaccurate mode, so the mode-shape error largely cancels, leaving a 2% discrepancy at 24 modes. The relative detuning results the designs are built on are therefore FE-accurate to a few percent; absolute frequencies of strongly steered designs should be read from FE or an enriched ($\gtrsim$ 24-mode) expansion. The paper's design evaluator uses 64 transverse modes, and the optimized design itself is evaluated on this FE model in Sec. 5.5 (detuning ratio 0.364 vs. 0.359 from the full SSM). A displacement-based von Kármán FE would replace the assumed modes throughout at higher cost, reduced by the same SSM.

field	model	f_1 [Hz]	γ	detuning @ a/h=0.4
UD [0] (homogeneous)	FE (28×21)	256.0	2.878×10^{10}	11.06%
	assumed-modes	256.3	2.842×10^{10}	10.96% (1%)
steered (strong)	FE (28×21)	334.4	2.515×10^{10}	5.73%
	assumed, 8 modes	358.9	2.546×10^{10}	5.30% (7%)
	assumed, 24 modes	345.7	2.584×10^{10}	5.65% (2%)

Table 5: Assumed-modes vs. FE backbone (parenthesized: detuning error vs. FE). "N modes" is the number of transverse modes in total (the N lowest), with 3×3 in-plane modes; the design evaluator uses 64 transverse ($m, n \leq 8$) and 6×6 in-plane modes (Table 1). This test therefore checks the transverse discretization and the nonlinear FE model on a smooth steered field (20°–65°); the in-plane resolution of the actual design is checked separately by the FE evaluation of the optimized field (ratio 0.364, Sec. 5.5). Homogeneous: the two agree to 1%, validating the FE nonlinear model. Strong steering: the frequency is a slowly-converging Ritz upper bound (+7% at 8 modes, +3% at 24), but the *normalized detuning* — the design quantity — matches FE to 7% at 8 modes and 2% at 24.

D Calibration against prior work, and SSM- vs EPMC-driven optimization

We reproduce two published results of [15] as a check of the reduction and its sensitivities, and compare SSM- and EPMC-driven optimization on one benchmark. *(i)* The backbone sensitivity of two coupled oscillators predicts the perturbed backbone to within about 3×10^{-5} in Ω over its soften-then-harden shape (the frequency first falls, then turns back up near an RMS amplitude of 1). Capturing the turn-back needs a high-order SSM; we use order seven, where [15] use order five with an analytic adjoint (Fig. 8). The damped eigenvalue $\lambda = -0.0191 \pm 0.6177i$ matches exactly. *(ii)* A clamped von Kármán beam is shaped by an initial curvature to a prescribed softening backbone, and full-order EPMC confirms the optimized SSM backbone to $< 0.1\%$ (Fig. 9). Two departures are noted. First, we use the axially condensed shallow-arch model rather than a three-DOF-per-node element. Second, we place the target amplitudes inside the cubic SSM's verified window. Eq. (33) of [15], taken literally, places a same-sign quadratic coupling on both masses, which is not energy-consistent; the potential-derived spring we use leaves the reported eigenvalue unchanged.

Validity window of the cubic SSM. Over the amplitudes reached by the full-order EPMC continuation ($a/h \leq 0.21$ for UD, ≤ 0.35 for the steered design; Fig. 6a), the cubic SSM backbone reproduces the EPMC detuning $\Omega/\omega_1 - 1$ to within 5%. The jump thresholds compared in Sec. 5.6 ($a_c/h = 0.208$ and 0.347) lie inside these ranges; we make no claim beyond them, where higher-order terms bend the true backbone.

EPMC as a design engine. The extended periodic motion concept [9] augments the equation of motion with a self-excitation that offsets damping, $\boldsymbol{M}\ddot{\boldsymbol{q}} + (\boldsymbol{C} - 2\delta\Omega\boldsymbol{M})\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} + \boldsymbol{f}_{nl} = \boldsymbol{0}$, and continues the harmonic-balance residual $\boldsymbol{R}(\boldsymbol{z}; a, \theta) = \boldsymbol{0}$ in amplitude. Its design sensitivity at a converged point follows from the implicit-function theorem with the Newton Jacobian already assembled, through one adjoint solve per amplitude, $\mathrm{d}\Omega/\mathrm{d}\theta = -\boldsymbol{w}^\top \partial_\theta \boldsymbol{R}$ with $\boldsymbol{J}_z^\top \boldsymbol{w} = \boldsymbol{e}_\Omega$. $\partial_\theta \boldsymbol{R}$ reuses the same $\partial\bar{\boldsymbol{Q}}/\partial\theta$ as the SSM gradient. The two engines are complementary. The SSM gives a smooth, analytic objective at a cost independent of amplitude and harmonic count, and is the search engine here. EPMC is exact at any amplitude it reaches, captures folds and internal

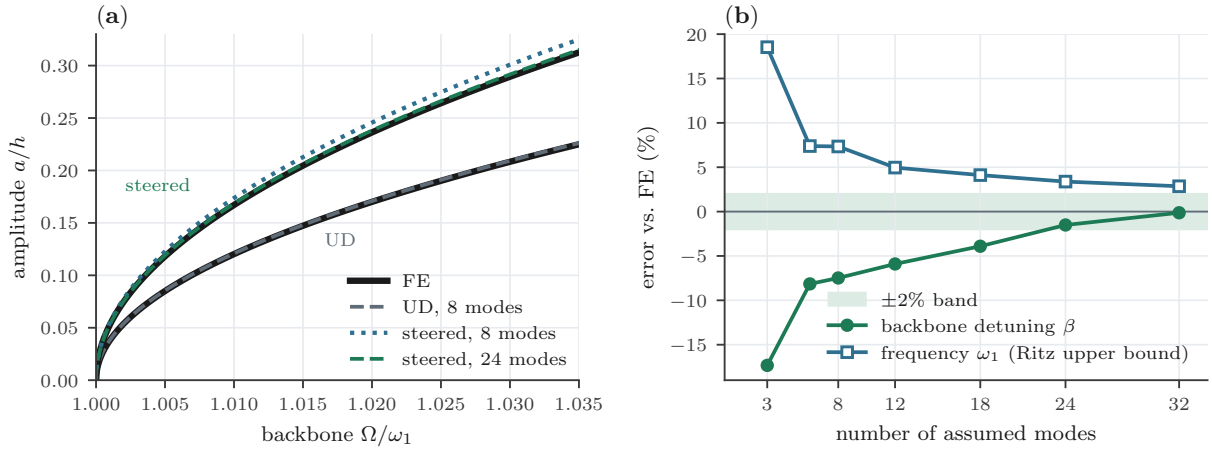


Figure 7: Finite-element validation of the assumed-modes nonlinear backbone. **(a)** Amplitude vs. backbone frequency (same orientation as Figs. 2 and 6); black: FE. For the homogeneous UD plate the 8-mode assumed-modes backbone (gray dashed) coincides with FE, validating the FE nonlinear model; for the strongly steered plate the assumed-modes backbone converges to FE as modes are added (8 modes dotted, 24 modes green dashed, overlying FE). **(b)** Ritz convergence toward FE (steered): the normalized backbone detuning enters the ±2% band by ~ 24 modes, whereas the absolute frequency is a slowly-converging Ritz upper bound. The design quantity (relative detuning) is the well-resolved one.

resonances, and serves as the validation of record. On the benchmark below, both find nearly the same optimum, and the SSM backbone is about two orders of magnitude cheaper per evaluation.

D.1 SSM- and EPMC-driven optimization on the same benchmark

To compare the two engines as *optimizers*, we solve the same design problem with each on the published von Kármán-beam backbone-shaping benchmark of [15] (their Sec. 5.2 / Fig. 3): a clamped–clamped beam whose backbone is shaped from hardening to a prescribed *softening* by an initial curvature $y = A_1 \sin(\pi x/L)$. The straight beam hardens (+2.2% relative detuning at RMS $= 0.2h_0$); the curvature injects a quadratic nonlinearity whose slaved response drives the SSM cubic coefficient negative, softening the backbone (it also stiffens the beam linearly, so the curvature-invariant measure is the detuning relative to each design's *own* ω_1). We design A_1 to hit a -1.5% softening target, driving the optimizer once with the SSM backbone and once with the full-order EPMC backbone. Both reach the target exactly. The optima are $A_1 = 8.03\,\mathrm{mm}$ (SSM) and $8.26\,\mathrm{mm}$ (EPMC), a 2.8% difference that is the cubic SSM's truncation error at RMS $= 0.2h_0$. The SSM backbone costs about two orders of magnitude less per evaluation (Fig. 10). This is the intended division of labor: the SSM adjoint does the search cheaply, and full-order EPMC confirms the optimum.

E Verification and reproducibility

Every computed result quoted in the main text and in Appendices A–C is a macro generated from the result files by `examples/paper_numbers.py` (`docs/paper/numbers.tex`), or a table written by a listed script; the few rounded bounds and estimates stated in words (e.g. "less than 0.02", the

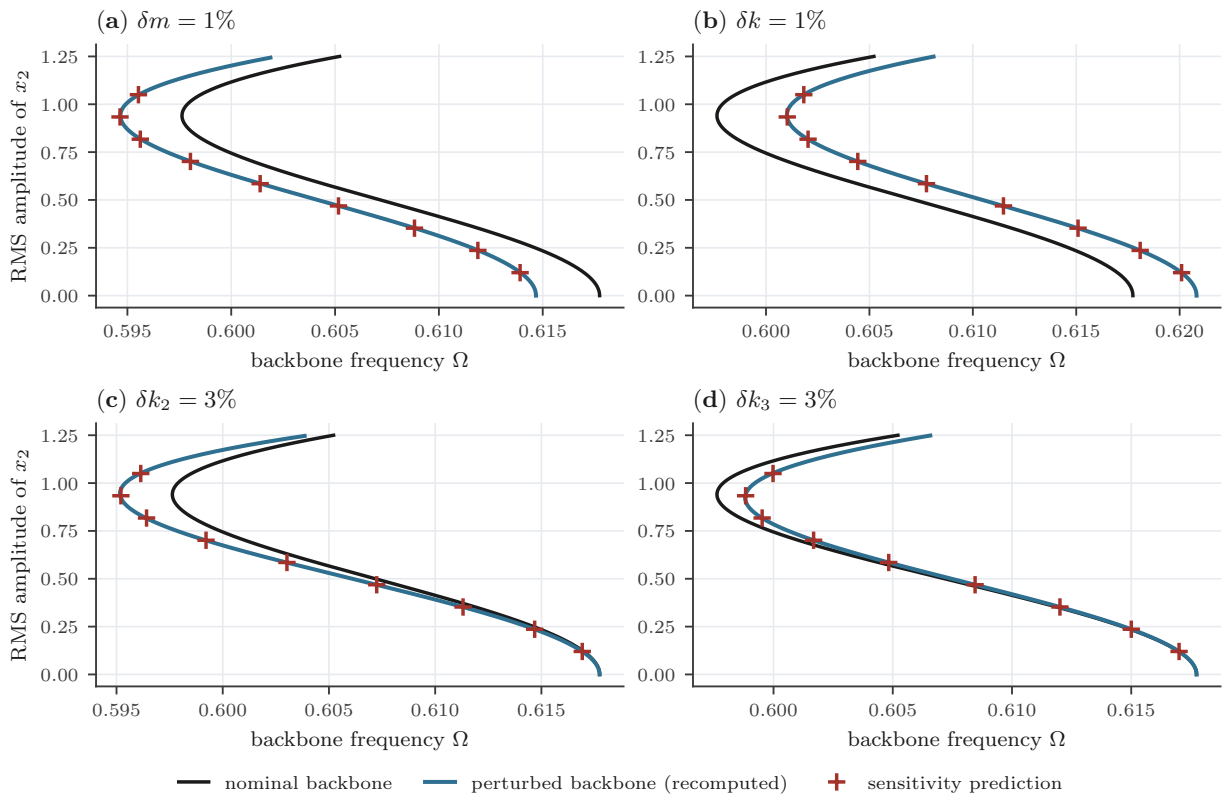


Figure 8: Calibration against [15], Fig. 2: backbone-sensitivity validation of two coupled oscillators. Each panel perturbs one parameter (m, k: 1%; k_2, k_3: 3%). Black: nominal backbone; blue: perturbed backbone, recomputed; red crosses: first-order sensitivity prediction from the nominal backbone. The predictions land on the recomputed backbones through the soften-then-harden turn-back, which the seventh-order SSM captures.

bang-bang estimate 0.4–0.5) follow from those values. The calibration values of Appendix D (the replicated benchmarks of [15]) are printed by the scripts listed for them below. Table 6 maps each result to the script that produces it. Unit tests are in `tests/` (`pytest`).

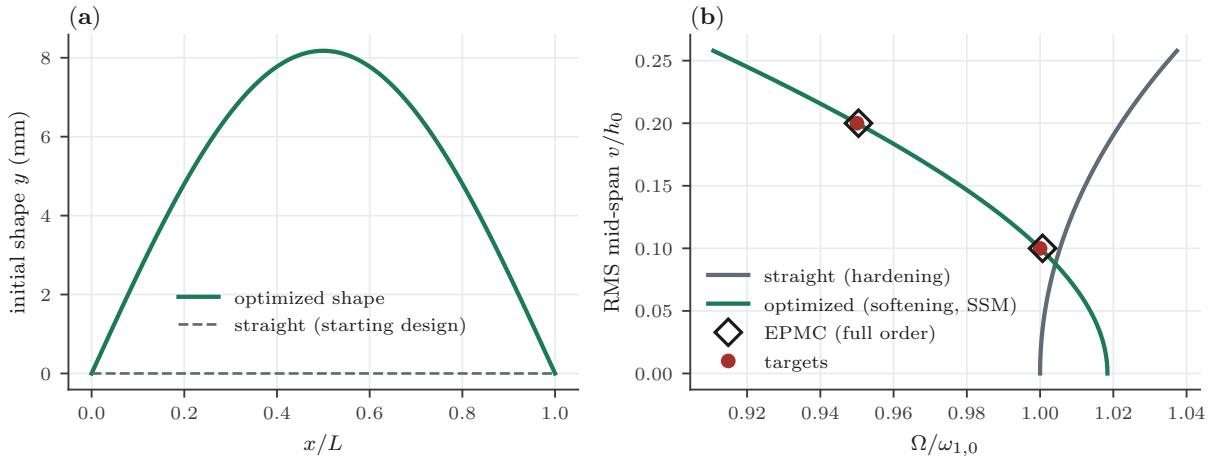
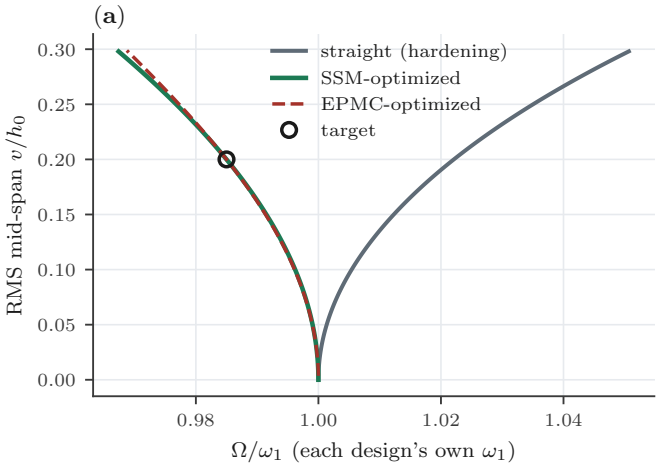


Figure 9: Calibration against [15], Fig. 3: a von Kármán beam shaped to a prescribed softening backbone. This problem shapes the beam with up to two harmonics (the optimum uses only the first) to pass through two prescribed backbone points; it is a different problem from that of Fig. 10, which tunes one amplitude to one target. **(a)** Optimized initial shape $y(x)$ and the straight starting design. **(b)** Backbones, normalized by the first frequency $\omega_{1,0}$ of the initial straight beam: the straight (hardening) beam becomes a curved (softening) one threading both target points; full-order EPMC (diamonds) confirms the optimized SSM backbone to $< 0.1\%$.



(b)

	SSM	EPMC
optimal A_1 (mm)	8.03	8.26
detuning at $0.2h_0$	-1.50%	-1.50%
backbone evaluations	12	10
wall time (s)	0.059	11.2
cost per backbone	$1\times$	$\approx 230\times$

Figure 10: Optimizing the same design with both frameworks (benchmark of [15]). Here one curvature amplitude A_1 is tuned to one softening target (a different problem from Fig. 9). **(a)** The straight beam's backbone hardens (leans right); the optimized initial curvature softens it (leans left) through the softening target; the SSM- and EPMC-optimized backbones nearly coincide. **(b)** The two optima and their cost; the SSM optimum differs from the full-order one by the cubic SSM's truncation error at RMS $= 0.2h_0$, at about two orders of magnitude lower cost per backbone evaluation (wall times are indicative only).

Table 6: Result → script map.

result	script (in `examples/` unless a path is given)
relaxed evaluator, gradient, tests	`nltopt/relaxed_plate.py`, `tests/test_relaxed_plate.py`
beam test (Sec. 5.1, Fig. 2)	`beam_symmetric_steering.py`, `fig_beam.py`
B-coupled beam (comparison)	`steered_beam_amplitude_robust.py`
achievable sets (Sec. 5.2, Fig. 3)	`levers_study.py`, `fig_levers.py`
mechanism, front, dose (Figs. 4, 5)	`mechanism_study.py`, `fig_mechanism.py`, `fig_limits.py`
resolution and pointwise ceiling	`ceiling_study.py`
full-order confirmation (Sec. 5.5), jump (Sec. 5.6)	`confirm_heldfreq.py` (SSM, thresholds, sweeps), then `epmc_check.py` (merges the EPMC backbones into the same file), `confirm_fine_sweep.py`, `fig_confirm.py`
independent FE check of the design (Table 3)	`fe_check_design.py`
basis convergence (Table 1)	`conv_table.py`
continuous-stack competitor (Sec. 5.2)	`stack_continuous.py`
frequency multiplier, linear spectra (Secs. 5.3, 6)	`mechanism_lambda.py`, `mode_ratios.py`
all quoted numbers and generated tables	`paper_numbers.py`
FE validation (App. C)	`plate_fe_backbone.py`, `plate_fe_fig.py`
calibration (App. D)	`coupled_osc_sensitivity.py`, `beam_backbone_opt.py`, `ssm_vs_epmc_opt.py`

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